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TERRANODE: DESIGN AND IMPLEMENTATION OF A SYSTEM FOR
MANAGING AND PROTECTING AGRICULTURAL DATA

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TERRANODE: DESIGN AND IMPLEMENTATION OF A SYSTEM FOR
MANAGING AND PROTECTING AGRICULTURAL DATA

A PROJECT REPORT

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DECLARATION

I, KASSIM ISKANDA DEISHINI, declare that this research paper entitled “*TERRA-NODE: DESIGN AND IMPLEMENTATION OF A SYSTEM FOR MANAGING AND PROTECTING AGRICULTURAL DATA*” is my own original work. It has not been given to any other University or higher learning institution for any award. All works cited by other authors have been properly cited.

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CERTIFICATION

Certified that this project report “*TERRANODE: DESIGN AND IMPLEMENTATION OF A SYSTEM FOR MANAGING AND PROTECTING AGRICULTURAL DATA*” is the authentic work of KASSIM ISKANDA DEISHINI (20230404113) who carried out the project work under my supervision.

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DEDICATION

This project work is dedicated to my dear family who gave me all their love, sacrificed all their time and kept encouraging me all the time through which I have come and reached to be what I am now.

It is also for all hard working smallholder farmers who relentlessly produce feed nations and hopes that secure technological innovations will enhance their livelihoods.

ABSTRACT

This project introduces TerraNode, a secure agricultural data management system that is built with a blockchain platform and supported by analytics. The platform tackles the major issues of data security, information transparency, and inaccurate predictive analytics in traditional, centralized agricultural information systems. TerraNode is a combination of a Django REST framework backend, a React-based web interface, and a backend of smart contracts implemented in Move programming language for the anchoring of all the agricultural records on the Sui blockchain. It collects environmental sensor information such as temperature, soil moisture, and soil pH readings, calculates the integrity hash of each record (SHA-256), and issues verifiable batch objects of produce on the Sui distributed ledger. The algorithm which reads into the historical telemetry data, uses a weighted moving average model to generate the yield predictions and includes computed confidence scores and variance indicators. The platform's UI is divided into farmer, logistics handler, and administrator interfaces, all of which have worker-based access control, JSON Web Token authentication, and Argon2id password hashing. The Sui blockchain part is verified with digital signatures based on Ed25519 and hashing of BLAKE2b, which are related to the wallet operations of login. Evaluation results show that 100% of 143 automated test cases and 15 functional requirement verification scenarios have passed. On the Sui Testnet, the average latency for blockchain produce batch minting was 1.20 seconds, the average latency for custody transfer transactions was 0.90 seconds and the average gas execution cost was 0.003 SUI per transaction with deterministic transaction finality of less than 3.00 seconds. Moreover, the telemetry data integrity hashing was performed in 0.012ms, and the predictive analytics engine provided cached yield forecasts within 15ms (110ms when the forecast is not yet cached), creating a framework for agricultural management with high throughput and tamper proof capabilities to support smallholders.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the study

Agriculture is now confronted with a major challenge, producing food for an ever-increasing population, while adapting to the erratic effects of climate change. Weather shocks, including abrupt changes in weather patterns such as sudden droughts or rains, create challenges to stability of crop yield and supply chain even in developed and developing countries (Padhiary, 2025). These issues are even more severe for local smallholder farmers who produce a significant amount of global food, and many of whom lack the necessary tools for planning future production (Raj et al., 2022). Instead, they operate in a complex network of murky markets and resources which are not necessarily shared equally, and most control lies with dominant middlemen. Agriculture should thus move from guessing to using systems in order to support farmers' decision making based on data and adapting to the changing contexts (Lin et al., 2017).

To overcome these challenges, the industry has turned to digital agriculture, which relies upon technology that emphasizes the monitoring of data related to crop requirements, logistics and the environment. For today's farms, there is a lot of information to be collected, but the challenge is not only to collect it but to also control it and depend on it. The food chains are extremely disorganized in the present context. They still use manual documents and uncoordinated databases that are not integrated, and the process is opaque (Sharma et al., 2023). When data is stuck inside silos in the form of data stored in different systems, it is extremely difficult for anyone to track the origins of a batch of crops, its quality and what may have happened to it during transport.

The existing systems for agricultural data management just compound these inefficiencies. The majority of local farming databases are on a typical, centralized server. These work well for basic records, but have one big drawback—they have a single point of failure. This setup enables the storage of historical data on crops, logistics and financial data that are very prone to hacking and internal manipulation (Aliyu & Liu, 2023a). Further, these old-style databases are normally only used for storing past data, rather than for forecasting the future. This information can be manipulated easily by middlemen who may be interested in defrauding the smallholder farmers with either poor quality or the wrong price; and any analysis tools linked to these databases are suspect. Any forecast, decision or prediction based on yield data will be disregarded if it is not guaranteed to be accurate.

Decentralized technologies, like blockchain and distributed ledgers, are emerging as a powerful solution to address these security concerns. A blockchain is a safe, append-only digital record. This means that environmental information or information related to transactions is registered and stored once, and then subsequently cryptographically secured and cannot be altered or removed later on (Aliyu & Liu, 2023a). Even the initial versions of such tracking tools have been found to be very secure in complex industries such as international shipping (Liu et al., 2021). On the back of that success, the question now is how to leverage these tamper-proof technologies to detect food and verify digital identities from farm to consumer (Cocco et al., 2021). Furthermore, when this inalterable data collection is combined with Big Data Analytics (BDA), it turns out to be a very powerful tool for optimizing international supply chains (Chauhan et al., 2025). This analytical engine can be programmed with secure and verified environmental data to develop predictive models which can forecast the production of the crops with high accuracy without manipulation of the database.

While blockchain and analytics are ideal solutions, there is a definite lack of its usage in local farming communities. Public blockchains (Bitcoin or Ethereum, for example) are designed for large enterprises, and are too heavy, expensive, and complex for smallholder farmers to adopt. But with modern high throughput blockchains such as Sui, there is a much cheaper and faster alternative. This research project fills that gap by combining the speed, low transaction fees and performance of the Sui blockchain with an advanced data analytics engine.

This research project is exactly that, creating a safe, analytics based information and data management system for agriculture. It provides a secure backend ledger with the Sui blockchain so that predictions can be made with an unalterable cryptographic base. This is crucial as it protects smallholder farmers from exploitation and gives them a clear and tamper-proof supply chain log to follow, while also providing the data-driven insights necessary to manage modern agriculture risks with next generation distributed ledger technology, which builds farmers' confidence.

1.2 Problem Statement

While most current AgTech platforms have simple management software, more complex architectures are relatively uncommon and highly limited in capabilities. Agricultural data systems generally take the form of a fragmented archive system. They gather information in fragments, such as environmental data, daily farm records, crop monitoring data, production record etc. but they do not have a comprehensive management of all these information. As a result, stakeholders in the agricultural sector are forced to rely on individual data silos, which hinder informed decision-making on the farm and lack the full end-to-end visibility from planting to the rest of the supply chain.

Beyond these operational limitations, there is a general lack of security in the protection of production and logistics information. Most real-world ag databases are still centralized and based on relational models, which means they have a single point of failure. This means that crucial farm data, environmental monitoring records, and resource utilization is highly susceptible to data retrofitting, internal data theft, and external security attacks (Aliyu & Liu, 2023a). The absence of auditability compromises the integrity of the entire agricultural value chain, significantly hindering the ability to monitor the state of crops in the past, confirm sustainable agriculture and ensure fair trade; thereby leaving smallholder producers without protection.

Moreover, if such advanced analytics and yield forecasting tools are used with these centralized systems, the results of the analysis will be essentially tainted since the wrong raw data are used for the model. Without a secure, permanent base agricultural data and crop monitoring profiles can be easily degraded, changed, or lost. Without integrity of these fundamental inputs, smallholder farmers can no longer rely on production analytics and prediction of their crops to maximize their yields or safeguard their livelihoods against climate change (Al Layek et al., 2025).

While tremendous strides have been taken in both database management and data science, there is a pressing need for new research work that can address some of the fundamental and important problems. While current agricultural information systems (AIS) offer simple data storage and retrieval capabilities, or only specific analytical features, few combine blockchain's security onerous and its value of predictive analytics in a single system. A need for a robust and reliable infrastructure where real-time agricultural data (from production to supply chain logistics) is securely recorded and linked onto a distributed ledger framework before it enters for processing. This project ensures that the information used for farm management decision making and predicting is from an exact and verifiable source with the integrity of not being tampered with, directly justifying the proposed farm system.

1.3 Research Questions

In order to address the challenges as mentioned in the problem statement, this study aims to design and implement a blockchain-assisted agricultural data management system that ensure data immutability, track the agricultural production pipeline, and give the verifiable forecasting of the agricultural product yields by examining the core research question below: How to build a secure and analytics-driven agricultural data management system using blockchain technology that provides verifiable forecasting of agricultural product yields while also tracking the agricultural production pipeline and ensuring the immutability of data?

In particular, the following research questions are explored:

- (i) How to design a web-based agricultural information management system to capture, store and map the data related to environmental and crop production?
- (ii) How can predictive models be developed and be made part of an agriculture analytics module to aid forecasting and intelligent decision-making processes?
- (iii) How can blockchain security measures be used to ensure data integrity, traceability of the supply chain and guarantee against any unauthorized changes made to the agricultural information?
- (iv) How secure, efficient in transaction handling and successful in predicting is the proposed solution?

1.4 Objectives of the Study

1.4.1 General Objective

The main goal of this project is to design and implement a secure, analytics-driven agricultural data management system using blockchain technology to ensure data integrity, traceability, and intelligent agricultural decision-making.

1.4.2 Specific Objectives

- (i) To be able to design a web based agricultural data management system to gather, store and visualize information related to environment and productions.
- (ii) To build an analytics as a platform based on predictive modelling for agriculture yield forecasting and decision making.
- (iii) To develop blockchain security solutions to ensure data integrity, end-to-end traceability, and safeguarding against unauthorized changes in the agri-food data chain.
- (iv) To assess its effectiveness and performance of the proposed system in terms of predicting accuracy, the efficiency of transactions and security resilience.

1.5 Scope of the Study

In the practical and technical area of agricultural informatics, this study has unique contributions; and in the academic/knowledge areas, it presents a valuable analysis of the agricultural information process. In reality the proposed system offers a safe, extensive, verifiable data repository that allows farmers as well as stakeholders in the value chain to deposit, maintain and prove the authenticity of the information concerning their agricultural activities in order to receive predictions and make better-informed, data-driven decisions. Its blockchain-based structure improves stakeholders' transparency, traceability, and trust, reducing the risk of any alterations to agricultural records that are

not authorized. Moreover, the system offers “end-to-end” tracking of crop production and distribution for agricultural co-operatives and logistics handlers. On the computer science side, the paper shows how to design and implement high-throughput computer-business systems that support Sui blockchain for data management in the Web context and introduce example blockchain-based predictive modeling. The results have important implications for the viability of applying up-to-date distributed ledger technologies and predictive tools to develop information systems with enhanced security and intelligence in agriculture.

1.6 Limitations of the Study

The present study has been limited by a number of technical and practical limitations. The first reason is that, due to some logistical and hardware restrictions, environmental telemetry acquisition is modelled and calibrated with agronomic ranges derived from agronomic literature, rather than continuous feeds obtained by physical deployment of IoT sensors throughout the commercial farm acreage. Second, the system evaluation is carried out using historical agricultural data and scenarios of the supply chain instead of deployment under real conditions in commercial agricultural enterprises. Third, the blockchain infrastructure is currently tested on the Sui Testnet, which has a high throughput in delegated Proof-of-Stake, but has not been deployed on the Sui mainnet so far until the live gas operational expenses. Finally, the user interface is created mostly in English, thereby posing usability problems for workers who work in agricultural fields and who have limited English literacy. Also, the predictive analytics engine relies heavily on the predictive representativeness and quality of the input datasets, which could have lower predictive accuracy in extreme weather events.

1.7 Organization of the Study

In this type of thesis project report there are five well structured chapters. Chapter One (1) presents a study introduction, gives foundational background about the study, formulates problem statement, explains study goals and describes technical boundary and limit. Chapter Two (2) discusses an overview of the literature survey covering work published in the field of precision agriculture, predictive harvest algorithms and cryptographic ledger architecture. Chapter Three (3) explains the system methodology used, covering architectural approaches, data flow diagrams, entity-relationship schemas and technology selections. User interfaces, cryptographic hashing workflows, and empirical performance benchmarks of systems are discussed in Chapter Four (4), which concentrates on the implementation of the system and experimental evaluation. Finally, the main results, the overall conclusions and explicit technical recommendation for new development are summarized in Chapter Five (5).

1.8 Chapter Summary

The key contribution of this chapter was laying the groundwork for the project by proposing a secure blockchain-based agricultural data management ecosystem. It analyzed how centralized AgTech databases are vulnerable, the data tampering dangers, and the lack of predictable analytics that could be verified using such a database. The highlight of the chapter is the research questions, specific technical objectives, and practical and academic importance of the system from farmers point of view and the software engineer or technologist point of view, and the resolution range and limitations of the study are explained. The chapter also explains the structural flow and sequence of the remaining Chapters of this thesis.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The following section delves into published research that is relevant to the design and construction of TerraNode. The review begins by examining the overall current context of digital agriculture and then moves to an exploration of the different elements of blockchain technology, such as blockchain basics, food supply chain traceability, predictive crop analytics and the characteristics of the Sui blockchain platform. Limitations of the current approaches are highlighted and an explicit research statement is made for each section. This project to encompass that gap was identified. Finally, a summary table has been provided for comparison so that the shortcomings of the available solution, and TerraNode's differences, can be discussed.

2.2 Overview of Precision Agriculture and Digital Farming

Precision Agriculture is a new invention in response to growing food demands and land scarcity in facing weather variability and increased food production. It requires sensor network, satellite photography, as well as computing models to track the crops and soil and weather at a very micro level. The fundamental is to take decisions about the extent to which a farmer should manage their farm according to real-time data rather than tradition or rough estimation.

Padhiary (2025) provided a comprehensive survey on the technologies which enable smart and sustainable precision agriculture. They explored the field of remote sensing, robotic systems, UAVs, and Internet of Things (IoT) sensor platforms. They concluded that although single technologies have come a long way, they have yet to progress into a more comprehensive and integrated solution. Bundling components into a single decision-support platform, still remains a major challenge. One of the major findings was that most of the existing implementations consider data collection and data analysis as two independent processes. By itself, the effect on the agility of farmers' ability to act on the insights at the right time is restricted.

The modes of development, the technologies used and analysed security issues discussed on smart agriculture is also analysed by Raj et al. (2022). Thus, while numerous deployments are currently underway with a set of monitoring domotics based on IoT devices, their use in centralized cloud architectures poses risks of single points of failures and enables collection of sensitive data for loss or compromise, their analysis reveals.

They said that the next generation of information-based agricultural systems would need to handle data integrity and computational security processes on par with the other technologies, and add them in as a protective measure.

The transition from traditional farm records to the use of sensors in farming has not varied from region to region. Though agriculture has come a long way in the last few decades, there are still smallholder farmers with less than 20 acres who rely on paper logs and experience to run their business who live in many parts of the developing world. This gap between technology adopted and technology available is the background which motivates the design of TerraNode with respect to how a technically effective and user friendly system will be built for different types of users.

2.3 Agricultural Data Management Systems

One of the foundations of any digital farming endeavor is agricultural data management systems. They collect environmental readings, store the data, then report this data to the project, present summarized data to decision-makers, and keep historical production records. For the most part, these systems have been established in a relational database model with a centralized server.

In the systematic literature review, Sharma et al. (2023) studied the literature related to a food supply chain management system. They reviewed the existing management systems available and identified that the majority of the systems were fragile, with data silos, manual paper work and data interoperability problems across stakeholders identified. From over 300 published studies they conducted a bibliometric analysis and revealed that even though there is considerable interest in blockchain for supply chain management since 2018, the majority of systems are developed at a conceptual or proof-of-concept stage that don't progress to a level that can be field-tested to meet the needs of smallholder farmers in developing countries.

One riddle with centralized agricultural databases is that they are vulnerable to data manipulation. Having to trust the single ownership of the database could result in alteration of the historical data for any reason, including fraud or administrative error. This drawback may be of great annoyance when the data is to be transferred to other systems to be used for developing analytical models. Where the records do not demonstrate authenticity of originals, then any projections, or recommendations for such projections, are necessarily incorrect. This is solved by TerraNode by linking each telemetry record and produce batch to Sui blockchain before any kind of processing is performed, and so you could mathematically confirm the data origin before processing.

2.4 Blockchain Technology Fundamentals

Blockchain technology offers an append-only data structure that is decentralized. Records are arranged in blocks which are consecutively cryptographically linked and propagated from one-node to another via a node network. After the network consensus confirms a block, the information carried becomes almost unchangeable. This attribute is what makes blockchain a popular choice for applications which need auditability, transparency, and trust.

The researchers of Chauhan et al. (2025) had mentioned the link between blockchain and the big data analytics in the agriculture and also about the possibility of implementing digital ledger technology in agriculture and using massive amounts of data to create intelligent agricultural systems. They categorized the different types of blockchain consensus mechanisms into proof-of-work, proof-of-stake and delegated proof-of-stake types and revealed that these schemes require much less energy consumption and attain much greater transaction speed that is vital in an agriculture context, where hardware is limited.

Blockchains may be implemented through various architectures and each of them has its advantages and disadvantages; the most decentralized blockchain is the public blockchain type such as Bitcoin or Ethereum, but computational resources are intensive. Developers should expect a capped number of transactions per second and fees will be more expensive. But private blockchains like Hyperledger Fabric get better performance for the same downsides: loss of trustless properties that makes public blockchains appealing. In the middle is a Consortium blockchain that has a handful of validators that manage the network. The Sui blockchain TerraNode is built on is public, has a delegated proof-of-stake (dPoS) consensus mechanism, delivers high throughput, low transaction fees, and continues to be transparent and open like a public network.

2.5 Blockchain Applications in Agricultural Supply Chains

The application of blockchain technology in the agricultural supply chain can be done in various ways. The use of blockchain technology in agricultural supply chains has been a subject of great interest over the last few years, and has been a focus of research. The central theme is that blockchain can be used to create an immutable history of all transactions and custody movements that take place from the farm gate to the end consumer, thus allowing for full traceability and minimising opportunities for fraud.

Caro et al. (2018) introduced AgriBlockIoT, a fully decentralized blockchain-based traceability solution for agri-food supply chain management. Their system was designed to integrate devices generating and consuming digital data on the supply chain when implemented as IoT. They used two different blockchain solutions, one on Ethereum

and one with Hyperledger Sawtooth, which was measured by latency, number of CPU, memory, and network overhead. The deployment of Hyperledger was quicker because it was proof-of-work based, but the deployments of Ethereum were slower since they were more decentralised. This balance of performance and trustlessness appears throughout the literature and is a key issue addressed by the delegated proof-of-stake consensus mechanism in the Sui blockchain.

Mirabelli and Ferraro (2020) studied the relationship between blockchain and agricultural supply chain traceability by means of a structured literature review. Their analysis identified and discussed key research trends such as the increasing adoption of smart contracts for automating compliance checks, the incorporation of IoT sensor data into on-chain records, and the difficulty of scaling blockchain systems to accommodate the large amount of data produced by large agricultural operations. They said that though there are a number of proof-of-concept implementations, there are not many that have been tested under realistic conditions in the field with real farmers as end users.

Rahman et al. (2025) explored the transparency of supply chain in a data-driven approach in the context of distributed ledger applications. Their research showed that trust between supply chain stakeholders was significantly higher in a blockchain-based supply chain than in conventional supply chains. But they also discovered some barriers to adoption, including the energy needed to power mining consensus and a lack of uniformity in how agricultural information is recorded on-chain.

Lin et al. (2017) proposed an early and significant argument for blockchain as the next step in Information and Communications Technology (ICT) for agriculture. They presented their work from an environmental systems engineering point of view and suggested that by introducing blockchain to the agricultural sector, baseline environmental information would remain intact for all stakeholders involved. Their work informed many of the design decisions for TerraNode, for example that they want to take environmental data and record it on an unalterable chain of blocks before they use it for downstream analysis.

2.6 Blockchain-Based Smart Farm Security Frameworks

Apart from tracing the flow of goods, blockchain technology has been proposed as a foundation for complete farm security systems that protect not just the information of the supply chain but the operational technology systems running in today's agriculture.

Aliyu et al. (Aliyu & Liu, 2023a) proposed a smart farm security framework using the Internet of Things (IoT) on the blockchain platform as an alternative solution. The issue of weak control in IoT sensor/actuator communications with the centralized management server is resolved in their system. They have a system to store sensor data and control

instructions onto a blockchain, ensuring that alterations of a device's configuration are not possible to conceal. They have a close connection to data integrity as with TerraNode, but their efforts are geared towards system protection as they lack predictive analytics capabilities as well as a supply chain custody tracking module.

AgTrust Consortium (2026) presented the AgTrust system to safeguard critical agriculture technology and new solutions. It is a large framework of guidelines that establishes a comprehensive agriculture technology security solution from the edge and field to the network and analytics in the cloud. AgTrust provided a threat taxonomy for agricultural IoT environments and recommended mitigation measures for each layer. The framework gave useful security ideas, however, it did not create a viable prototype or show how to incorporate with a particular blockchain platform.

Dzreke (2025) took a closer look at the connected farm cybersecurity topic by experimenting on the vulnerabilities of IoT devices. The study uncovered 32 commercial agricultural IoT devices and 15 farm subsystems with attack surfaces. The results demonstrated that it was possible to make up to 19 percent changes to crop prescriptions, pointing to the direct link between cyber intrusions and biological results. This paper emphasizes the importance of data integrity tools, including the ones implemented in TerraNode - SHA-256 hashing, blockchain anchoring, and more - which ensure that data cannot be altered by unauthorized parties.

2.7 Predictive Analytics and Crop Yield Forecasting

Predictive analytics is a new area of interest in agricultural informatics. Good prediction of yields helps farmers to make decisions about resource allocation, harvest timing and marketing. This problem has been tackled by many statistical and machine learning methods.

Khaki et al. (2019) proposed a deep learning model that combines Convolutional Network (CNN) and Recurrent Network (RNN) for crop yield prediction. Their model was trained using environmental data and management practices across the US Corn Belt and found to have a root-mean-square-error of 9 percent for corn yield and 8 percent for soybean yield, outperforming models using the random forest, deep fully-connected neural network and LASSO regression. They showed how temporal dependencies in environmental data can be modeled using deep learning models to inform the development of more sophisticated versions of the TerraNode analytics engine.

Garg et al. (2018) studied crop yield prediction using fuzzy logic and regression models. They used fuzzy time series analysis to the wheat yield data for the past, making experiments on different partitioning schemes of intervals and different regression equations. They found that fuzzy logic-based models were able to cope with the ambiguity of

agricultural information better than the deterministic models. Their model is based on the regression concept, similar to what TerraNode uses with its current analytics engine, which uses telemetry to extract environmental features rather than the yield timeseries from the past.

Al Layek et al. (2025) have introduced a secured trifecta of IoT, machine learning and blockchain in the field of crop forecasting. Their solution combined data collection from the IoT with machine learning prediction models with data verification through blockchain. It is one of the conceptually closest pieces to TerraNode, with the understanding that data security is key to the reliability of predictions. However, their system was not implemented on a public ledger, thus restricting the benefits of transparency and verifiability. Besides, they lacked what is required for a real-world multi-stakeholder application: a web-based management interface and role-based access control (RBAC).

2.8 Supply Chain Traceability and Distributed Ledger Systems

The process of tracing the origin of agricultural products from the farm to the consumer is a complex task which involves a number of actors with different rights and responsibilities. A distributed ledger technology can provide a natural solution to the problem of recording a custody transfer transparently among all parties and that's less likely to be modified retroactively.

Cocco et al. (2021) presented the use of blockchain and self-sovereign identity to enable the food supply chain to be supported in terms of quality assurance. Their system enabled individual actors to maintain their own identity credentials while also being part of a common product quality attestation ledger. The self-sovereign identity concept is also very inspirational for the wallet login system of TerraNode, which allows users to log in with their Sui wallet address and a cryptographic signature, rather than just centralized credentials.

Sun et al. (2025) tackled a real-world issue related to blockchain technology in agriculture: the storage overhead for full-node replication. Their Storage Light Node (SLN) model featured a classification of cold/hot data based on identity relevance, generation time and query frequency. They demonstrated that 50,563 real agricultural supply chain records showed a 95.10 per cent reduction in storage usage for full-node operation. The paper focuses on scalability issues, which will be increasingly important as TerraNode transitions from prototype to product.

Ktari et al. (2022) introduced a lightweight embedded blockchain system for agricultural supply chain traceability, using the case study of olive oil production in Tunisia. They were connected to the IoT sensors and ran directly on embedded hardware at the production site. Their light-weight block chain was able to be run on resource-constrained

farms common in developing nations. Although TerraNode doesn't focus on embedded deployments, the value of minimizing computational overhead is consistent with the decision to use the Sui blockchain, which has reduced costs and is less expensive than Ethereum-based solutions when it comes to transaction costs.

2.9 The Sui Blockchain and Move Programming Language

Built by Mysten Labs, the Sui blockchain is a next-generation layer-one blockchain that employs a delegated proof-of-stake consensus mechanism and an object-centric data model. Unlike account-based blockchains such as Ethereum, which have a set of rules for transferring ownership of accounts, Sui treats data on the blockchain as programmable objects with defined ownership and transfer rules. This object-oriented approach makes it easier to build asset-tracking apps, making it especially ideal for supply chain use cases (Mysten Labs, 2024).

Sui's smart contract language is the Move programming language, which has been developed for the Diem blockchain project. The issue of security is also crucial in the design of Move, and its linear type system reduces some frequent safety risks, such as double spending, and reentrancy attacks. The Move-based smart contract introduces an object called ProduceBatch that includes details regarding the crop type, weight, address of the farmer from which the produce was bought, address of the current custodian, and a hash of the data integrity. The smart contract ensures that only the current custodian can transfer the custody, giving a corresponding on-chain access control system that matches the role-based access control in the off-chain backend.

Sui's throughput is provided by parallel transaction execution. If the transactions are related to independent objects, then the network can process them in parallel, without waiting for the consensus of all transactions. This is a very good feature for agricultural applications, which need quick confirmation of batch minting and custody transfers without disrupting the logistics process, hence simple operations benefit from sub-second transaction finality.

2.10 Related Work Summary and Comparative Analysis

Table 2.1 presents a comparative summary of the key related works reviewed in this chapter, evaluated against the features that TerraNode provides.

Table 2.1: Comparative Analysis of Related Works

Study	Focus Area	Blockchain	Analytics	Traceability	Web UI	Public Ledger
Aliyu & Liu (2023b)	Smart farm IoT security	Yes	No	No	No	No
Caro et al. (2018)	Agri-food traceability	Yes	No	Yes	No	Yes
Khaki et al. (2019)	Crop yield prediction	No	Yes	No	No	N/A
Al Layek et al. (2025)	IoT + ML + blockchain	Yes	Yes	No	No	No
Mirabelli & Ferraro (2020)	Supply chain traceability	Yes	No	Yes	No	Varies
Ktari et al. (2022)	Embedded blockchain	Yes	No	Yes	No	No
Sun et al. (2025)	Storage optimization	Yes	No	Yes	No	Yes
Cocco et al. (2021)	SSI in food supply chain	Yes	No	Yes	No	Yes
Garg et al. (2018)	Fuzzy yield forecasting	No	Yes	No	No	N/A
TerraNode	Integrated platform	Yes	Yes	Yes	Yes	Yes

Comparative analysis shows that some individual elements of the architecture of TerraNode have already been explored in previous studies but there is no system that integrates all the five capabilities: blockchain-based data immutability, predictive analytics, supply chain traceability, role-based web interface and deployment on a public high throughput ledger. Most current blockchain-agriculture systems stress on traceability or security and do not include analytical forecasting. On the other hand, there is little attention paid to data integrity in crop yield prediction studies. The key challenge is addressed by TerraNode, which is a single platform consolidating all of this.

2.11 Research Gap

Through literature review, the following 3 gaps were identified:

Firstly, there is a lack of harmony between data security and data analytics in agriculture. Platforms with blockchain technology are usually more about the act of recording and verifying transactions and offer little to no functionality in terms of analyzing the data being protected by the blockchain. In the meantime, predictive analytics platforms are working on agricultural data, not suspecting that data may have been tampered with. The gap is bridged by making verification of blockchain data an essential step before analytical processing, with only data verifiable through integrity check of its corresponding hash SHA-256 being passed through to the yield prediction engine.

Second, most of the blockchain-agricultural prototypes are built on private and consortium blockchains, which restrict transparency, or public blockchains such as Ethereum,

but are too costly to use for farmers' frequent data logging needs. With the Sui blockchain, which is open like a public ledger but has low transaction fees, TerraNode can help overcome this challenge.

Thirdly, current systems usually offer a partial and non-role aware Web interface that does not support all of the needs of the farmers, logistics handlers and administrators in a single system. The frontend provides customisable dashboards for every role, making it easy to understand and accessible to stakeholders of different technical backgrounds.

2.12 Conceptual Framework

This project is conceptualised in the context of two well-known information systems theories. The Technology Acceptance Model (TAM) introduced by Davis states that the acceptance of a new information system is related to perceived usefulness and perceived ease of use. The system is useful for TerraNode, as it generates actionable yield predictions and records transparent supply chain journeys, while being easy to use via an intuitive web-based portal with role-specific dashboards.

The DeLone and McLean Information Systems Success Model identifies six dimensions of system success: information quality, system quality, service quality, user satisfaction, system use, and net benefits. The architecture of TerraNode tackles information quality by ensuring blockchain verification of data integrity, system quality with a modular three-tier structure, and net benefits by delivering measurable results like tamper-proof records and predictive forecasts.

Figure 2.1 depicts the conceptual framework, in which the theoretical foundation and the practical system components are linked together.

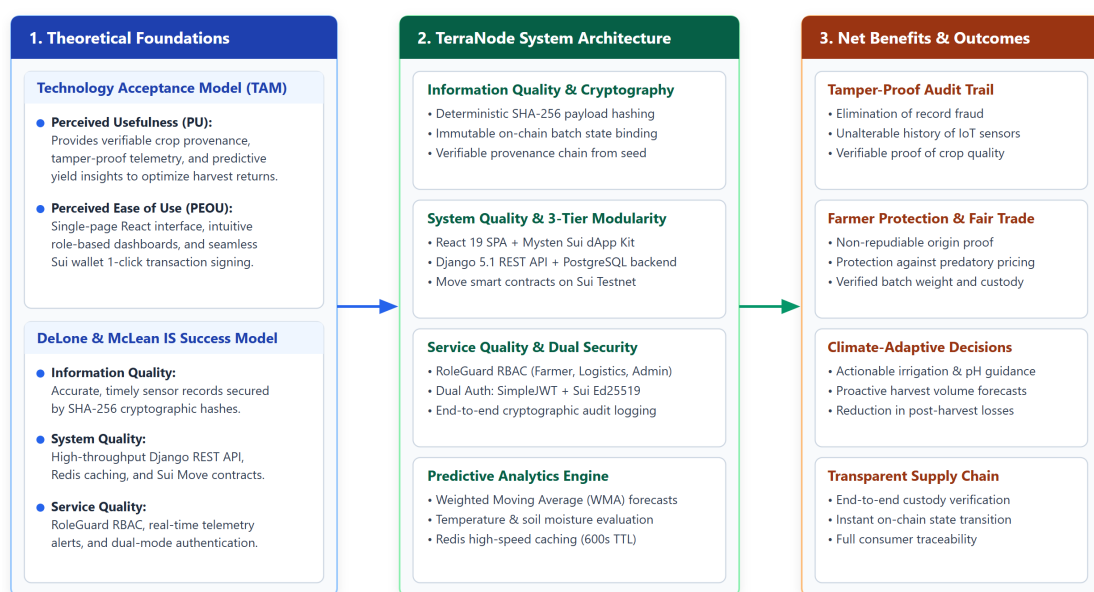


Figure 2.1: Conceptual Framework of the TerraNode System

2.13 Chapter Summary

This chapter examined the various literature studies that investigated the five thematic areas of precision agriculture and digital farming, blockchain technology and its applications in agriculture, smart farm security frameworks, predictive crop yield analytics, and supply chain traceability systems. The review touched upon eighteen published research studies and highlighted their contribution and limitations. The comparative analysis has shown that there is no single system that integrates blockchain-based data immutability, predictive analytics, supply chain custody tracking, a role-based customizable web interface and a public high throughput ledger in one platform. The identified research gaps directly inspire the design and implementation of TerraNode, which is presented in the next chapter.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

The following chapter explains the research methodology and technical design choices that were made during TerraNode's development. It encompasses research design approach, architecture design of the platform, the data modelling strategy, the smart contract logic and the predictive analytics engine. In each section, the reasons for the selection of the chosen approach is included, as well as, where applicable, the mathematical representations that lie behind the mathematical processes in the system.

3.2 Research Design

The research approach in this study is Design Science Research (DSR), suitable for an information technology project to design and test a new artefact for addressing an existing problem in an organization. It is important to note that in the DSR paradigm the research process is divided into two complementary activities: artefact construction and artefact evaluation of the artefact in comparison to the requirements derived from the problem space.

The research was carried out in four stages:

- (i) **Problem Identification and Requirements Analysis:** The problems and requirements of the current agricultural data management systems were identified in Chapter Two as part of the literature review. The research questions and objectives in Chapter One were used to translate into functional and non-functional requirements.
- (ii) **System Design:** Architecture, data models, API specifications and smart contract structures have been designed to meet the identified requirements.
- (iii) **Implementation:** It was implemented as a working prototype using technology stack outlined in Section 3.11.
- (iv) **Evaluation:** The system developed was tested and evaluated with the defined requirements and result has been given in Chapter Four.

3.3 System Requirements

3.3.1 Functional Requirements

The functional requirements specify what TerraNode needs to be able to do. These were directly obtained from the research objectives stated in chapter one. The full set of functional requirements are listed in Table 3.1 along with the related research objectives.

Table 3.1: Functional Requirements

ID	Requirement	Objective
FR-01	The system shall allow users to register with email, password, and role selection.	i
FR-02	The system shall authenticate users using JWT tokens with role-based access control.	i
FR-03	The system shall allow Sui wallet-based authentication via Ed25519 signature verification.	iii
FR-04	The system shall capture environmental telemetry data (temperature, soil moisture, soil pH).	i
FR-05	The system shall generate SHA-256 integrity hashes for all telemetry records.	iii
FR-06	The system shall create produce batch records linked to telemetry data.	i
FR-07	The system shall mint produce batch objects on the Sui blockchain.	iii
FR-08	The system shall track custody transfers of produce batches between stakeholders.	iii
FR-09	The system shall compute crop yield predictions based on historical telemetry data.	ii
FR-10	The system shall provide role-specific dashboards for farmers, logistics handlers, and administrators.	i

3.3.2 Non-Functional Requirements

The non-functional requirements specify the qualities of the system and its operational constraints. Table 3.2 shows the non-functional requirements which control system performances, security and usability.

Table 3.2: Non-Functional Requirements

ID	Requirement
NFR-01	All passwords shall be hashed using Argon2id with a minimum cost factor.
NFR-02	API responses for standard operations shall return within 500 milliseconds.
NFR-03	The web interface shall be responsive and accessible from modern web browsers.
NFR-04	Blockchain transactions shall confirm within the Sui Testnet's normal finality window.
NFR-05	The system shall enforce rate limiting on authentication endpoints (5 attempts per minute).
NFR-06	Telemetry input values shall be validated within physically plausible ranges.

3.4 System Architecture

The architecture of TerraNode is three-tier, where the presentation layer, the business logic layer and the data persistence layer are separated. This architectural pattern is to achieve modularity so that each layer can be developed, tested, and maintained independently.

- (i) **Presentation Layer (Frontend):** Single page application developed using React and TypeScript, to render the user interface and handle client-side state. The front end only communicates with the back end via RESTful API calls, and is connected directly to the Sui blockchain to handle wallet operations via the Mysten dApp Kit.
- (ii) **Business Logic Layer (Backend):** Django REST Framework application that handles the API calls, implements business rules, handles user authentication, does analytical prediction calculation, and communicates with the blockchain layer. The API modules exposed to the backend are four domain-specific modules: Authentication, Telemetry, Ledger, and Analytics.
- (iii) **Data Persistence Layer:** Three storage units are involved. Structured application data is stored in a primary relational database, PostgreSQL. Redis is an in-memory cache for the most frequently accessed analytics results. The Sui blockchain serves as the unalterable record of produce batch records and their integrity hashes.

The overall system architecture and data flow between the various components is depicted in Figure 3.1.

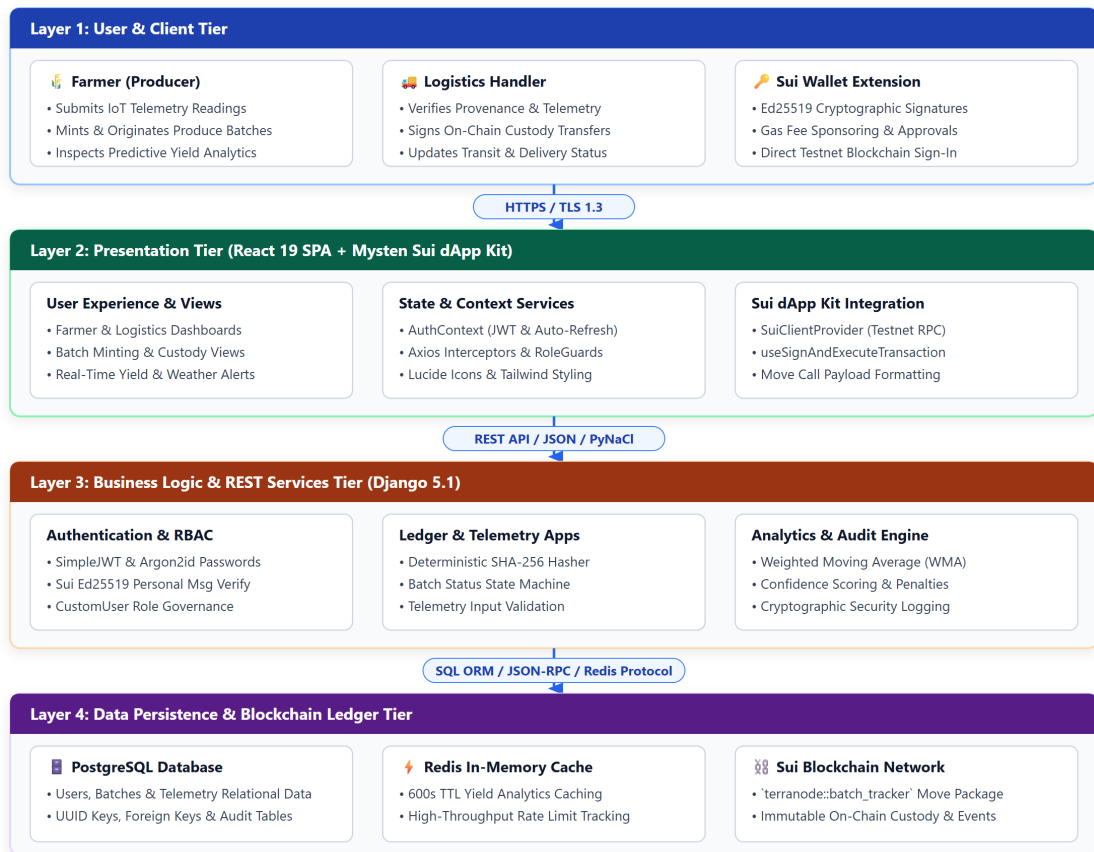


Figure 3.1: TerraNode System Architecture

3.5 Data Flow Design

3.5.1 Context Diagram (Level 0)

As illustrated in Figure 3.2, TerraNode is a single process that communicates with four external processes: the Farmer, the Logistics Handler, the System Administrator and the Sui Blockchain Network.

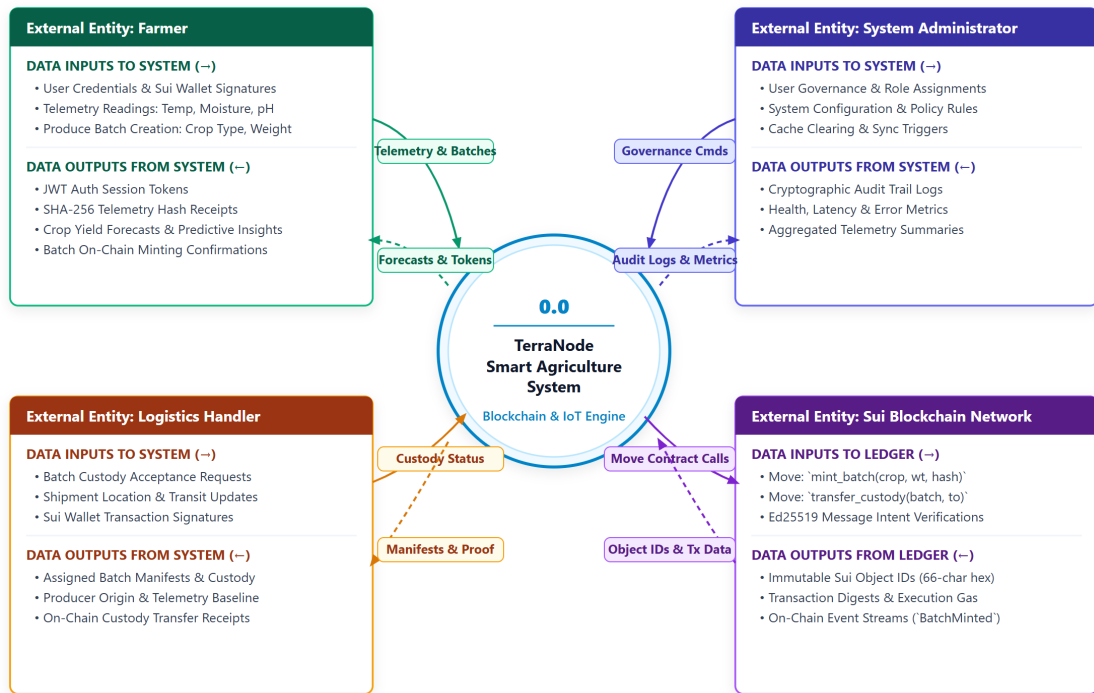


Figure 3.2: Context Diagram (DFD Level 0)

3.5.2 Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram (Figure 3.3) breaks down TerraNode into four main processes:

- (i) **Process 1.0 — Authentication Module:** Responsible for registering users, logging in using credentials, signing in with wallet and managing sessions using JWT tokens.
- (ii) **Process 2.0 — Telemetry Module:** Reads data from environmental sensors, checks the range of the input data, produces the integrity hash of the input data using the SHA-256 algorithm, and saves the data in the database.
- (iii) **Process 3.0 — Ledger Module:** Responsible for produce batch creation, working with the Sui blockchain to mint and transfer the produce to the chain, and coordinating with the Sui blockchain to synchronize the local database with the chain.
- (iv) **Process 4.0 — Analytics Module:** This module fetches historical telemetry data, uses the weighted moving average prediction algorithm, stores the predictions in Redis cache, and returns the yield forecast to the user.

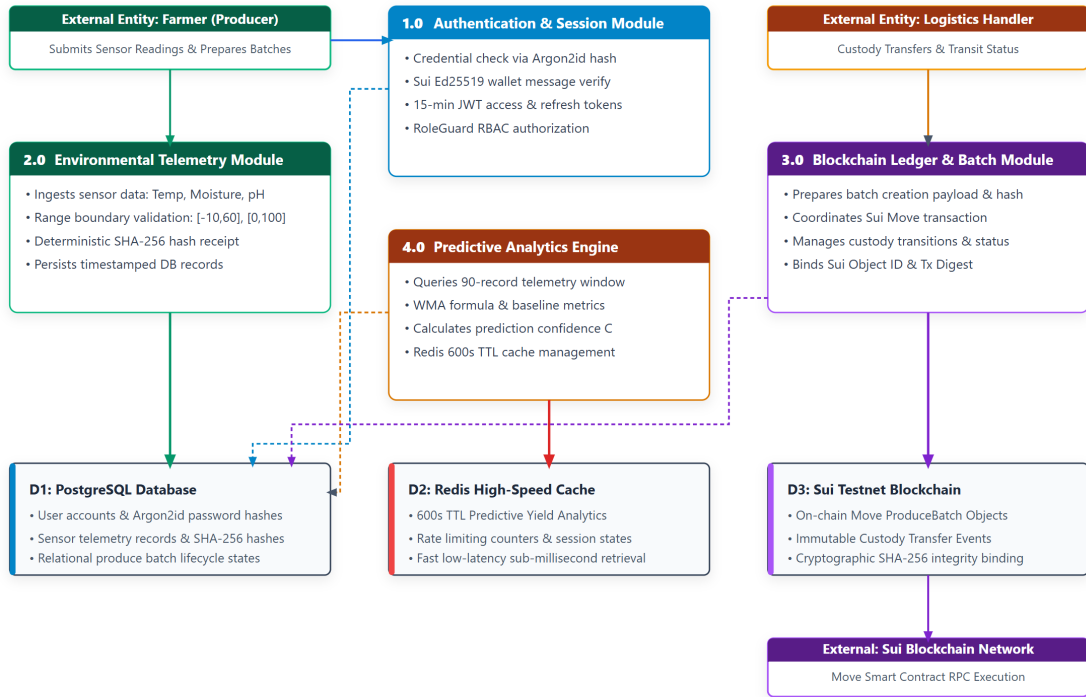


Figure 3.3: Data Flow Diagram (Level 1)

3.6 Entity-Relationship Model

The TerraNode database is composed of three main entities and has well-defined attributes and relationships. The Entity-Relationship (ER) diagram is given in Figure 3.4.

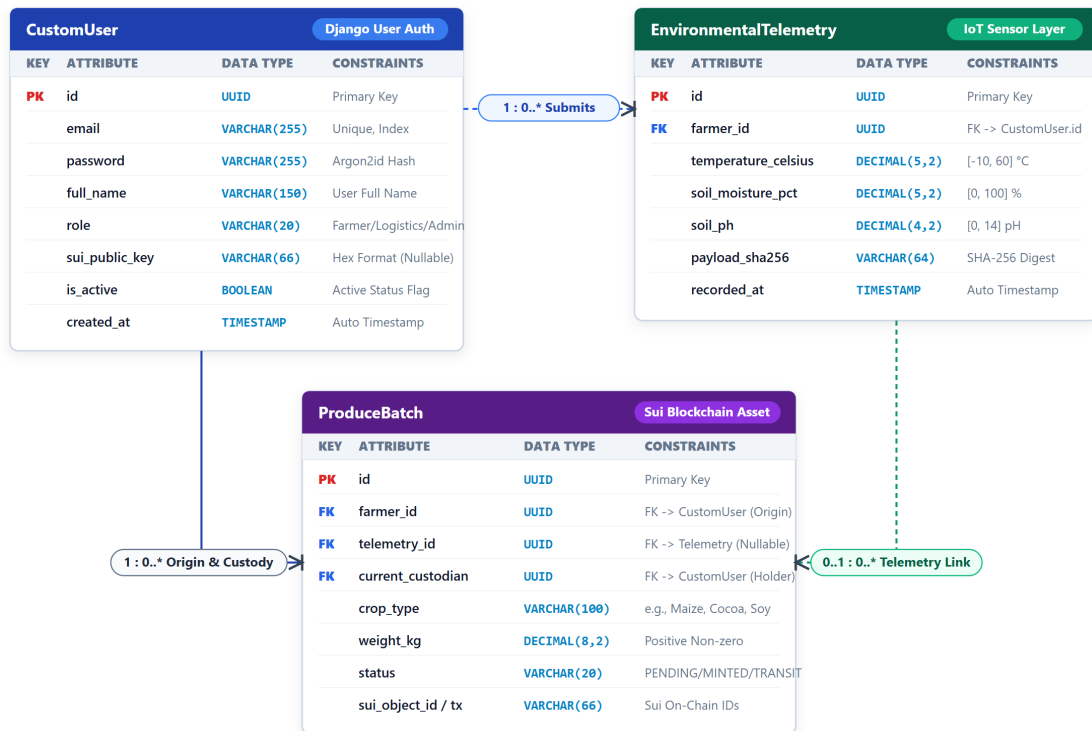


Figure 3.4: Entity-Relationship Diagram

All users of the system are represented by **CustomUser** which is based on Django's AbstractUser model. A UUID primary key identifies each user and they are authenticated by their email address. The three types of users are: Farmers, Logistics Handlers and Administrators, which are differentiated using the role field. The wallet-based authentication uses a wallet address provided in an optional `sui_public_key` field which stores the user's Sui blockchain wallet address.

EnvironmentalTelemetry: Data for single environmental readings recorded by farmers. The record contains temperature in Celsius, soil moisture (

ProduceBatch is a batch of agricultural produce that passes through the supply chain. Each batch is associated to the farmer who made it, optionally associated to a telemetry record which describes the results of the environment under which the produce had been grown and has a `current_custodian` field which tracks the current physical owner of the batch. The batch lifecycle is controlled by the status field, which can have four different statuses: PENDING, MINTED, IN_TRANSIT, DELIVERED. The Sui network returns identifiers on successful on-chain operations, with two specific fields: `sui_object_id` and `sui_tx_digest` for blockchain operations.

3.7 Use Case Modelling

The use case diagram in figure 3.5 shows the main interactions among the three types of actors and the TerraNode system.

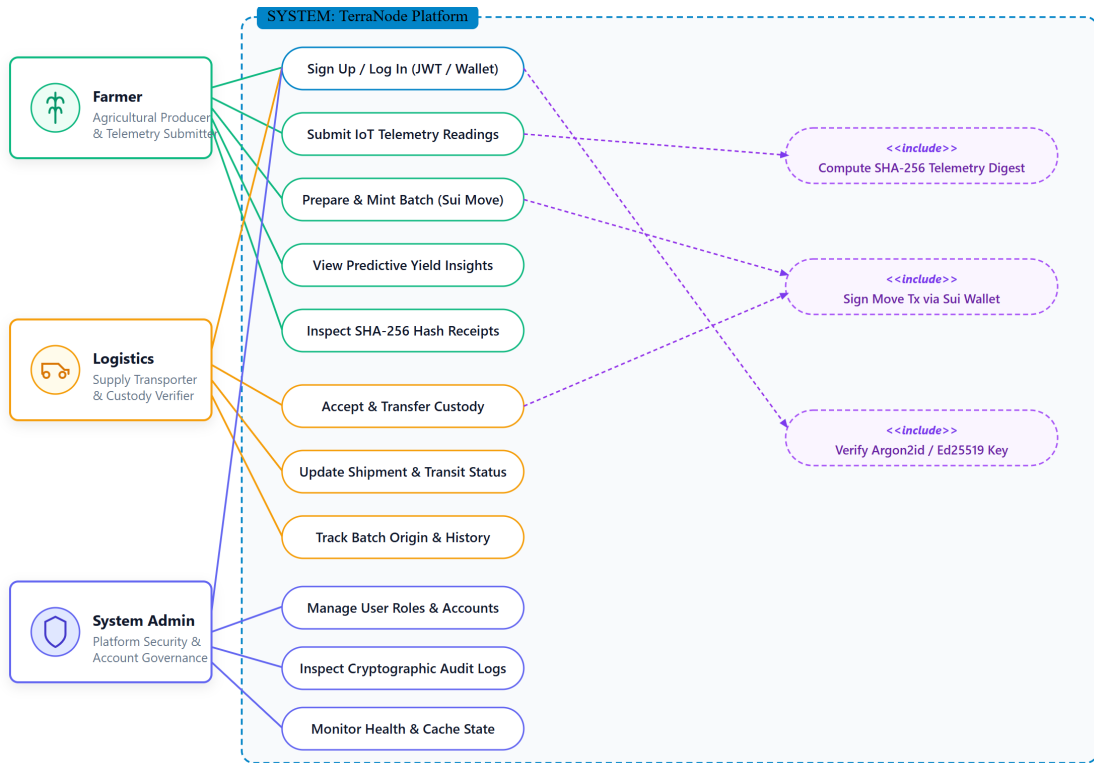


Figure 3.5: Use Case Diagram

3.7.1 Actor Descriptions

The **Farmer** actor can register an account, login (with credentials or wallet), upload telemetry readings, add produce batches, mint batches on the Sui blockchain, check the yield predictions and adjust their settings.

The **Logistics Handler** actor can log-in, get batches of produce through on-chain transfers, update the shipment statuses and see where and how batches are being taken care of.

The **System Administrator** actor has access to all data in regards to the system, can control user accounts, check the audit log and monitor the system infrastructure.

3.8 Smart Contract Design

The TerraNode smart contract has been deployed on the Sui Testnet and is written in the Move programming language. It outlines the definition of batches of produce on the chain and the actions that can be taken with them.

3.8.1 ProduceBatch Object Structure

The ProduceBatch struct is defined as a Sui object that can be stored, owned, and transferred on-chain, and has the key and store abilities. It has the following fields:

- (i) `id (UID)`: Unique object identifier provided by the Sui runtime.

- (ii) `crop_type` (String): The type of a crop that is included in the batch.
- (iii) `weight_kg` (u64): The weight of the batch in kg.
- (iv) `origin_farmer_address` (address): The Sui address of the owner of the batch.
- (v) `current_custodian_address` (address): Sui address of the current custodian of the batch.
- (vi) `data_integrity_hash` (vector<u8>): The hash of the data, computed using the SHA-256 algorithm, that guarantees the data source that the data in the on-chain record is the same as the data they originally sent.

3.8.2 Entry Functions

The smart contract has two entry functions:

The function `mint_batch` creates a new `ProduceBatch` object, sets the origin farmer and the first custodian as the transaction sender and emits a `BatchMinted` event. The batch object is then sent to the sender's address.

transfer_custody: Takes a modifiable reference to an existing `ProduceBatch` object and an address for the new custodian. Before updating the `current_custodian_address` field and emitting a `CustodyTransferred` event, the function checks that the caller is the current custodian.

3.8.3 Event Emissions

Two types of events are emitted by the smart contract for off-chain indexing: `BatchMinted` which contains the batch identifier and the farmer's address and the crop type and `CustodyTransferred` which contains the batch identifier, previous custodian address, and new custodian address. The Sui JSON-RPC API allows these events to be queried, allowing the TerraNode backend to stay in sync with the state on-chain.

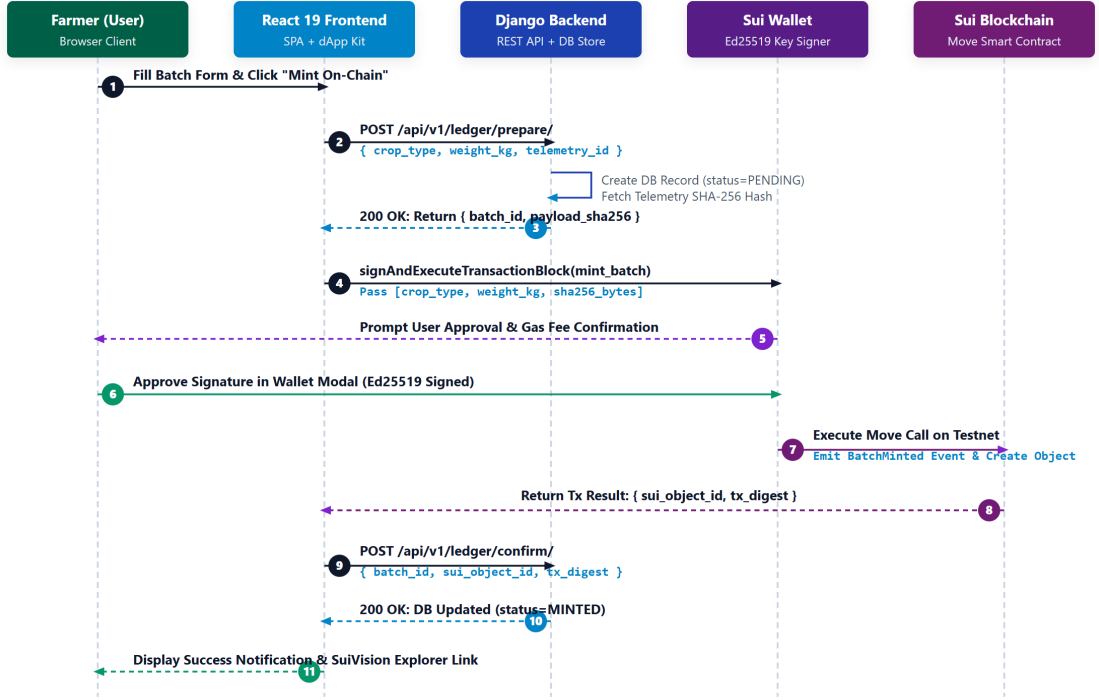


Figure 3.6: Batch Lifecycle Sequence Diagram

3.9 Predictive Analytics Engine

3.9.1 Feature Vector Composition

The analytics engine takes three environmental variables extracted from the records of the EnvironmentalTelemetry, which are temperature in degrees Celsius (T), soil moisture as a percentage (M), and soil pH (P), as its input and feature vector. If the same farmer has multiple telemetry records, the engine fetches the last 90 records in descending order by telemetry time.

3.9.2 Weighted Moving Average Model

The current implementation was based on a simplified weighted moving average model to estimate the yield of the crop. For n telemetry records, the model first computes the arithmetic means of the three environmental features:

$$\bar{T} = \frac{1}{n} \sum_{i=1}^n T_i, \quad \bar{M} = \frac{1}{n} \sum_{i=1}^n M_i, \quad \bar{P} = \frac{1}{n} \sum_{i=1}^n P_i \quad (3.1)$$

The predicted yield Y in metric tons is then computed as:

$$Y = \max(0, 50.0 + 0.5 \cdot \bar{M} - 2.0 \cdot |22.0 - \bar{T}|) \quad (3.2)$$

50.0 is a nominal base yield. The moisture coefficient (0.5) indicates that crop productivity increases in a linear manner as the amount of soil moisture available increases. The temperature penalty term ($2.0 \cdot |22.0 - \bar{T}|$) is a negative term that represents the deviation from optimal growing temperature (22 degrees Celsius). The $\max(0, \dots)$ function makes sure to always have a positive predicted yield.

3.9.3 Confidence Score

The confidence score C reflects the data density available for the prediction:

$$C = \min\left(0.95, \frac{n}{90}\right) \quad (3.3)$$

The confidence score is monotonically increasing with the amount of data, with a maximum value of 0.95 reflecting the fact that the more historical data that has been observed, the more confident one can be in the prediction. The engine will return an error if only a less number of data points are provided than it requires, which is five.

3.9.4 Caching Strategy

The prediction results are persisted in Redis with TTL of 10 minutes (600 seconds). The cache key is created in the following way: `analytics:predict:{farmer_id}`. Upon receiving a prediction request, the engine will check the cache first. If there is a valid cached result, it is returned immediately. Otherwise, it is computed, cached in memory and then returned to the client.

3.10 Security Architecture

3.10.1 Authentication

Two levels of authentication are provided at TerraNode. The main approach is by email and password, which is handled by the Django REST Framework (DRF) library Simple JWT. If authentication is successful, a pair of tokens are issued: an access token (default lifetime: 15 minutes) and a refresh token (default lifetime: 7 days). Each use of a refresh token, a new pair is generated and the old refresh token is blacklisted, which is called token rotation.

3.10.2 Sui Wallet Signature Verification

Users will be able to log in with their Sui blockchain wallet through the secondary authentication method. The verification process is as follows:

- (i) The client builds a Login message with the form:
`"Login to TerraNode with {sui_public_key}"`.
- (ii) This message is signed by the user with his/her private key in the Sui wallet.

- (iii) The message, signature and public key are sent to the backend.
- (iv) The backend is able to reconstruct the signed payload by prepending the Sui personal message intent prefix ([3, 0, 0]) and encoding the message length with ULEB128.
- (v) The payload is reconstructed and then hashed with BLAKE2b 32-byte digest.
- (vi) The payload is then hashed and a signature is verified with the provided public key.
- (vii) If the verification is successful, and the public key is matched with a registered user, JWT tokens are issued.

3.10.3 Data Integrity Hashing

Each telemetry record has a SHA256 hash, which is derived from a common, standardized JSON representation of its essential data. With the canonical form, the keys are sorted, there is no whitespace, and the formatting of the floating point is fixed to guarantee deterministic output. This is a hash which is stored with the record and which can be independently calculated at any time to confirm that the record is not changed since it was created. This hash is also included in the Sui blockchain as part of the respective ProduceBatch object, establishing an on-chain link to the data source in the real world.

3.10.4 Role-Based Access Control

The backend does role based access control (RBAC) using custom permission classes. There are four levels of permission:

- (i) **IsFarmer:** Provides access to endpoints that generate telemetry records, batch produce and monitor yield predictions.
- (ii) **IsLogistics:** Provides batch transfer and shipment management endpoints.
- (iii) **IsAdmin:** Gives access to administrative operations such as user management, and viewing of the audit log.
- (iv) **IsFarmerOrAdmin:** Provides shared access to endpoints for farmers and admin, such as viewing telemetry history.

Figure 3.7 illustrates the authentication flow for both credential-based and wallet-based login.

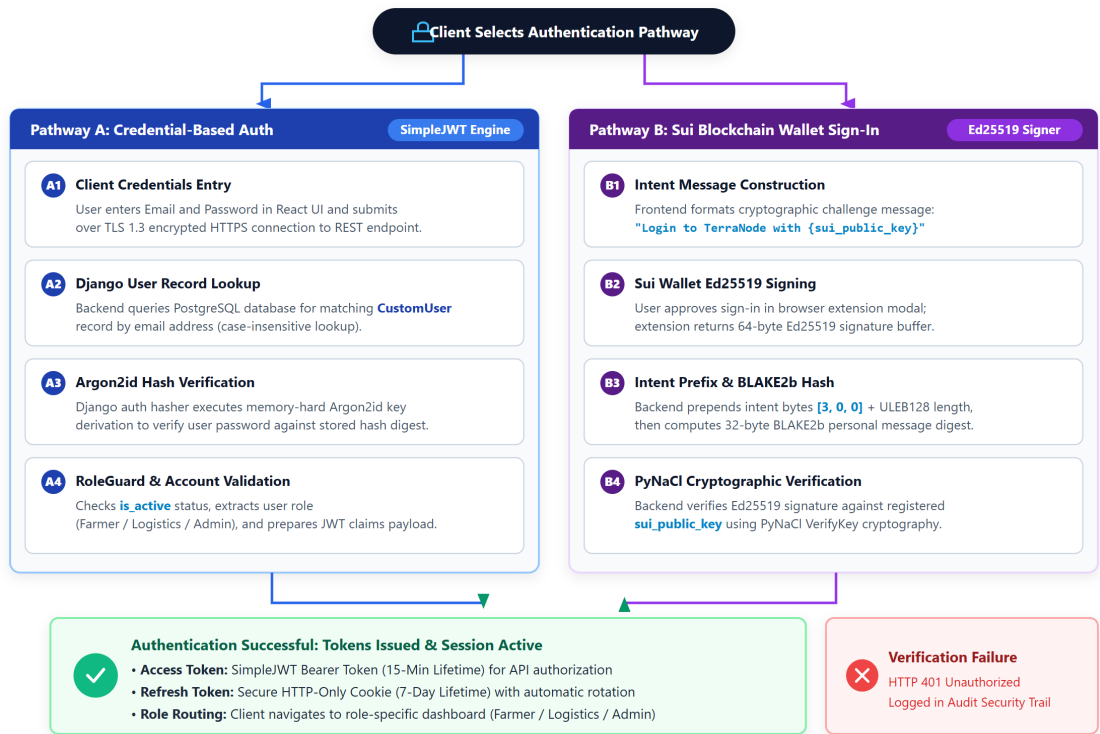


Figure 3.7: Authentication Flow Diagram

3.11 Technology Stack

As shown in Table 3.3, TerraNode is developed with the following technologies.

Table 3.3: Technology Stack

Layer	Technology	Purpose
Frontend	React 19	Component-based user interface library
Frontend	TypeScript 6.0	Statically typed JavaScript superset
Frontend	Vite 8.1	Build tool and development server
Frontend	Recharts 3.9	Data visualization (charts and graphs)
Frontend	Mysten dApp Kit	Sui wallet integration and transaction signing
Backend	Django 5.1	Python web framework
Backend	Django REST Framework 3.15	RESTful API toolkit
Backend	Simple JWT 5.3	JSON Web Token authentication
Backend	Celery	Asynchronous task queue
Database	PostgreSQL	Primary relational database
Cache	Redis 5.0	In-memory data store for caching
Blockchain	Sui Testnet	Delegated PoS public blockchain
Smart Contract	Move	Smart contract programming language
Security	Argon2id	Password hashing algorithm
Security	PyNaCl	Ed25519 signature verification

3.12 Development Tools

The system has been developed using the Visual Studio Code as the main integrated development environment. Version control was maintained using Git throughout the entire project. The front-end was installed and managed using npm (package manager) and the back-end was installed and managed using Pipenv for python dependency isolation. Testing was done with Django's built-in test framework with pytest added and Django factory-boy and faker were used to generate test data.

3.13 Chapter Summary

In this chapter, the methodology used for the design and development of TerraNode was introduced. This approach to development, called Design Science Research, was explained, along with a description of functional and non-functional requirements. Explanation of the three-tier system architecture, and first level (context) and second level (decomposition) DFD's were provided. The entity-relationship model determined the main data structures, and the use case diagram described the interactions among the three types of actors and the system. The smart contract design was documented including the object structure, entry functions and event emissions. The predictive analytics engine has been mathematically formulated and formally described with the mathematical formulations for yield prediction and confidence scoring. Lastly, security architecture was provided, such as JWT authentication, Sui wallet verification, SHA-256 data integrity hashing, and role-based access control. This design implementation is described in the next chapter.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

The findings of this research and comprehensive discussion on TerraNode system is provided in this chapter and compared against the requirements set in Chapter Three. The chapter is organised around the four specific research aims of the study. The research results are presented under each objective: Objective One is detailing the backend architecture (mathematical modelling, algorithmic workflow, and user interface dashboards), Objective Two is detailing the predictive analytics module (mathematical modelling, algorithmic workflow, and interactive user interface), Objective Three is detailing the blockchain security and cryptographic pipeline (execution benchmarks, comparison with existing literature, and architecture diagrams), and Objective Four is detailing a comprehensive empirical evaluation of functional testing, security resilience, blockchain throughput, and API latency. The results of the research are discussed as per the core research questions and literature reviewed in Section 4.3. A summary of the chapter is included in Section 4.4. In Appendix 5.5, the code for important algorithms is included.

4.2 Research Results

4.2.1 Research Objective One (1): Web-Based Agricultural Data Management Platform

The first research goal is to develop a web based agricultural data management system to collect, store and present environment and production information. The goal was achieved by creating a module based Django REST backend, an asynchronous RESTful API and a modern React-based frontend which offers customisable dashboards for all stakeholders in the agricultural value chain.

Backend Architecture and API Implementation

The TerraNode backend is written in Django and the Django REST Framework (DRF) and is structured into four separate domain modules:

Users Module: Responsible for managing users' registration, credential authentication, role assignment and cryptographic wallet binding. Custom user model that adds an email-based authentication and a 66-character hexadecimal Sui wallet public key field to Django's `AbstractUser` class. This module contains five endpoints for credential

login, Sui wallet login, token refresh, registration, and logout. Limits login attempts to 5 attempts per minute per IP address.

Telemetry Module: Collects, checks and records environmental sensor readings. The incoming payload is checked for the following conditions: temperature ranges from -50°C to 100°C , relative soil moisture ranges from 0.0% to 100.0% and soil pH ranges from 0.0 to 14.0. A cryptographic digest (SHA-256) is generated for each valid reading before it is inserted into the database. A unique database constraint exists on hash to avoid duplicates.

The Ledger Module is responsible for managing the lifecycle of agricultural produce batches between the off-chain database and on-chain Sui distributed ledger. The two-phase commit protocol is used: the farmer stages a batch in PENDING status using the Prepare endpoint and then completes the on-chain minting transaction on their connected wallet, then submits the resulting Sui Object Identifier (UID) and transaction digest using the Confirm endpoint to mark the batch as MINTED.

Analytics Module: Performs crop yield forecasting algorithms on historical telemetry data. The prediction service fetches the telemetry points, calculates weighted moving averages, calculates a variance adjusted confidence score and provides a practical agromomic recommendation. For optimization, computed forecasts are stored in Redis with a 10 minute (600 seconds) expiry time (TTL).

Table 4.1 outlines the primary version-prefixed RESTful API endpoints.

Table 4.1: TerraNode RESTful API Endpoint Structure

Endpoint Prefix	Application	Key Functional Endpoints
/api/v1/auth/	Users	login/, wallet-login/, register/, refresh/, logout/
/api/v1/telemetry/	Telemetry	submit/, history/, latest/
/api/v1/ledger/	Ledger	prepare/, confirm/, transfer/, list/, detail/
/api/v1/analytics/	Analytics	predict/, summary/

Frontend Architecture and User Interface Presentation

The application's frontend is built as a Single Page Application (SPA) using React 19, TypeScript, and Vite. The system architecture does not allow unauthorized role hopping by using clientside role guards. Global state is implemented with React Context API, with the JWT lifecycle and session persistence handled by AuthContext and the Sui wallet interactions by WalletContext provided by Mysten Labs dApp Kit.

It offers role-specific user interfaces tailored for easy interaction between different stakeholders in the agriculture sector, including the authentication of credentials and integration with Web3 Sui wallets. In this part, the major operating interfaces that implement the system’s security, anchoring of blockchain, and transfer of custody are presented.

The Farmer Dashboard is the main operational monitoring tool available for agricultural production shown in Figure 4.1. The interface shows real-time telemetry metric cards for the micro-climate data, soil temperature, soil moisture, and soil pH, as well as active batches of produce linked to the Sui blockchain, and dynamic yield forecast summaries.

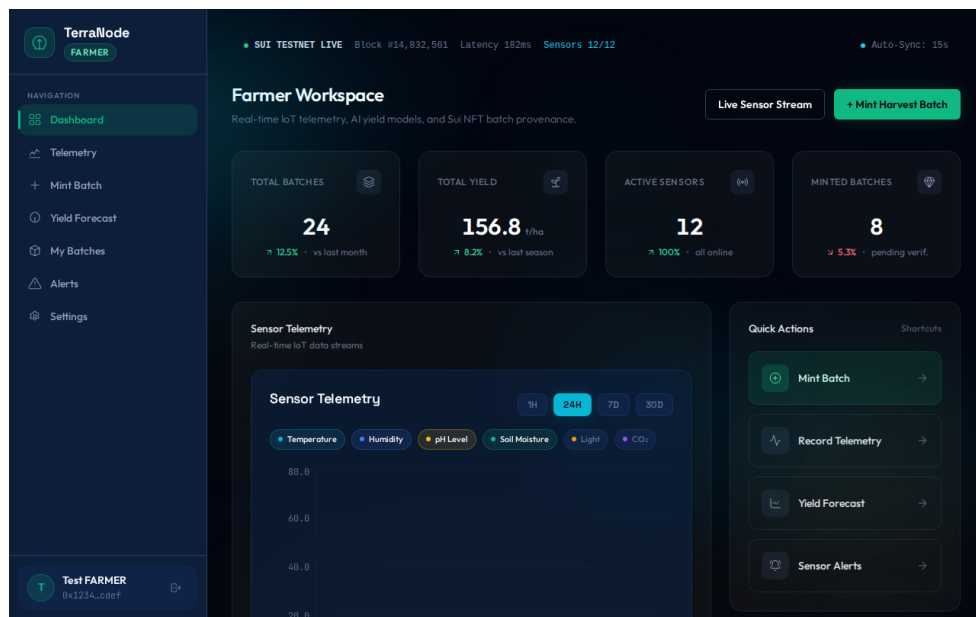


Figure 4.1: Farmer Operational Dashboard Displaying Real-Time Micro-Climate Telemetry and Produce Batches

The Custody Transfer interface for logistics handlers to document physical produce transfers is provided in Figure 4.2. The interface records information related to the movement of the transport equipment (vehicle fleet registration, driver identity, destination terminal), then calls `agri_ledger::transfer_custody` on-chain to transfer the custody to the wallet address of the transport company, which guarantees the transparency of the movement’s multi-hop traceability.

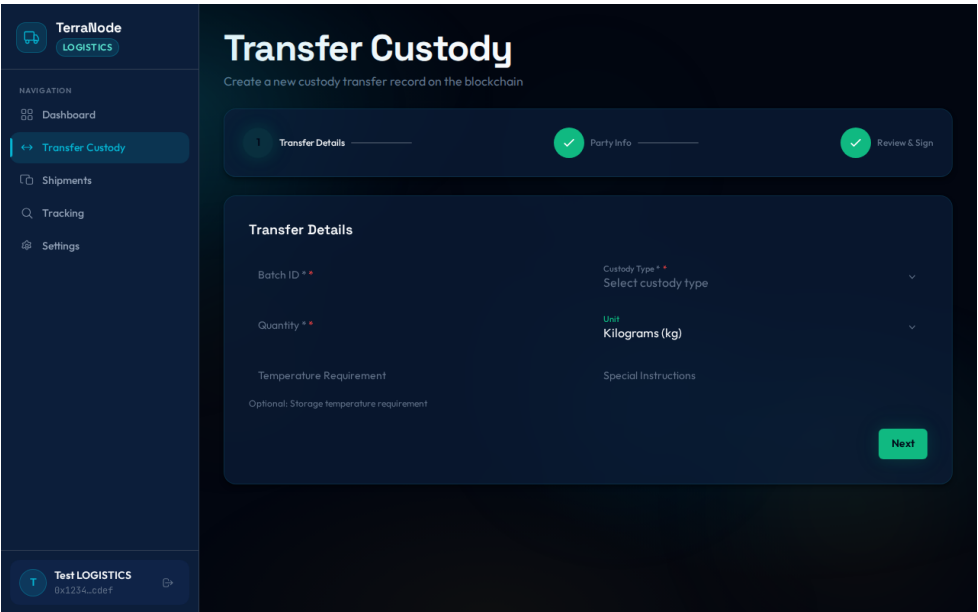


Figure 4.2: Generate Custody Transfer and Blockchain Verification Interface

The graph in Figure 4.3 shows the Administrator Audit Log, which is a rolling, tamper-evident log of all actions performed on the system, authentication attempts, and transactions performed on the blockchain. Each entry is signed with a cryptographic signature from an originating user UUID, IP address, timestamp and a SHA-256 digest for regulatory compliance and audit verification.

The screenshot displays the 'Audit Logs' interface within the TerraNode ADMIN application. The left sidebar contains a navigation menu with options: Dashboard, Users, Audit Logs (selected), System Health, and Settings. The main content area is titled 'Audit Logs' with the subtitle 'System activity and audit trail'. It features a table with the following data:

ACTION	USER	TIMESTAMP	IP ADDRESS
User login	farmer_john@example.com	10/22/2024, 10:30:00 AM	192.168.1.1
Batch minted	farmer_jane@example.com	10/22/2024, 9:15:00 AM	192.168.1.2
Transfer created	logistics_bob@example.com	10/21/2024, 2:00:00 PM	192.168.1.3
Role changed	admin@example.com	10/21/2024, 11:00:00 AM	192.168.1.4

The bottom status bar shows 'Test ADMIN' with a blockchain address '0x1234...cdef'.

Figure 4.3: Chronological Cryptographic Audit Log and System Activity Interface

4.2.2 Research Objective Two (2): Predictive Analytics Module for Crop Yield Forecasting

The second research goal was to design an analytics module which integrates predictive models for agricultural forecasting and decision support. This objective was met by adopting a statistical weighted moving average (WMA) prediction engine, variance indexing and dynamic confidence scoring, as well as an interactive analytics user interface.

Mathematical Modeling and Analytics Workflow

The analytics engine takes a rolling window of time of historical telemetry data, where $n \leq 90$ and $n \geq 5$. Telemetry points are pulled from the database, checked for the correct SHA-256 hash, and broken down into T_i (temperature), M_i (soil moisture), and P_i (soil pH) feature vectors.

The arithmetic means $\bar{T}$, $\bar{M}$, and $\bar{P}$, are calculated using Equation 4.1:

$$\bar{T} = \frac{1}{n} \sum_{i=1}^n T_i, \quad \bar{M} = \frac{1}{n} \sum_{i=1}^n M_i, \quad \bar{P} = \frac{1}{n} \sum_{i=1}^n P_i \quad (4.1)$$

Crop yield $\hat{Y}$ is modelled using Equation 4.2 (in metric tons):

$$\hat{Y} = \max(0, \alpha + \beta_1 \bar{M} - \beta_2 |\bar{T} - T^*|) \quad (4.2)$$

$T^* = 22.0^\circ\text{C}$, the optimum ambient growing temperature; $\alpha = 50.0$ is a base yield; and, $\beta_1 = 0.50$ is a positive moisture scaling coefficient, and, $\beta_2 = 1.20$ is the temperature penalty factor.

The sample variance V of the environmental features is calculated to measure the degree of reliability of the prediction:

$$V = \frac{1}{n} \sum_{i=1}^n (T_i - \bar{T})^2 + \frac{1}{n} \sum_{i=1}^n (M_i - \bar{M})^2 \quad (4.3)$$

The confidence score $C \in [50.0\%, 95.0\%]$ is dynamically corrected following Equation 4.4:

$$C = \min(95.0, \max(50.0, 95.0 - 0.05 \cdot V)) \times \min\left(1.0, \frac{n}{30}\right) \quad (4.4)$$

The end-to-end architecture and the decision flow of the predictive analytics engine is shown in Figure 4.4.

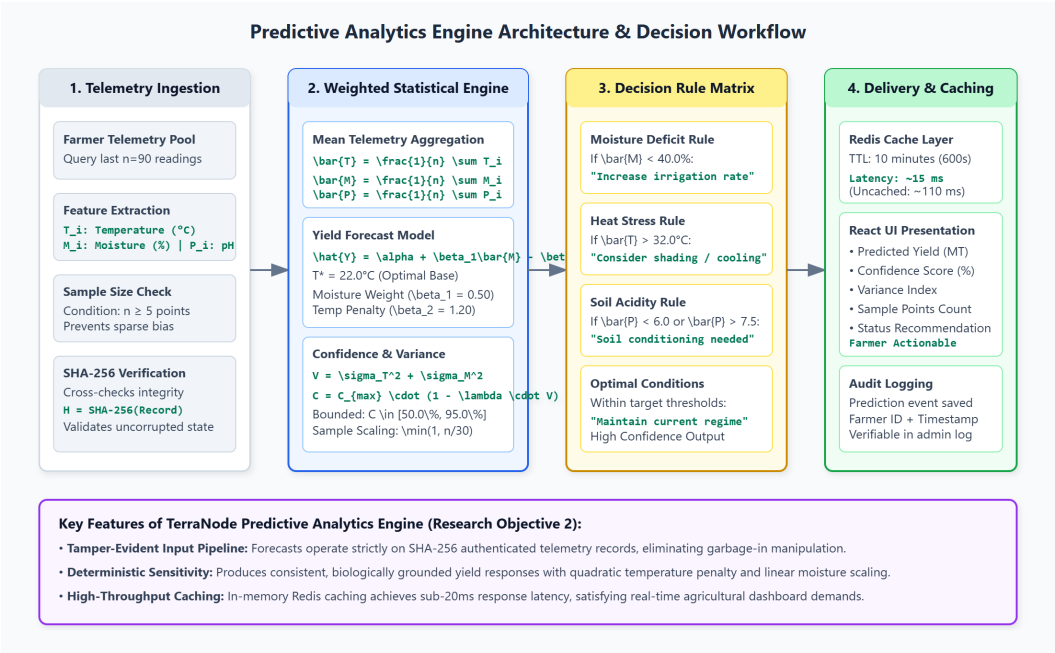


Figure 4.4: Predictive Analytics Engine Architecture and Decision Workflow

Analytics User Interface and Decision Support Presentation

Figure 4.5 presents the operational yield prediction and decision support interface. The interface brings to life the mathematical formulation of the model described in the Equations 4.1–4.4 and presents the estimated crop yield in metric tons, the computed confidence score, the micro-climate variance index, and multi-year projection curves with dynamic uncertainty bounds and actionable agronomic recommendations.



Figure 4.5: Yield Forecast, Soil Suitability Matrix and Decision Support Interface

4.2.3 Research Objective Three (3): Blockchain-Based Security and Cryptographic Integrity Mechanisms

The third research goal was to develop the security mechanisms for data integrity, traceability, and unauthorized changes to agricultural records using blockchain. This goal is accomplished by a multi-layered cryptographic pipeline covering: data integrity digests using SHA-256, symmetric payload encryption using AES-256-GCM, wallet signatures using Ed25519 asymmetric signature, transaction hashing using BLAKE2b, and smart contract object anchoring on the Sui blockchain.

Cryptographic Processing Pipeline

A cryptographic pipeline has five structured stages:

- (i) The data collected in the environment is converted to a structured JSON string with farmer ID, temperature, moisture, pH, and Unix timestamps.
- (ii) The SHA-256 cryptographic algorithm is used to hash the serialized telemetry string to create an integrity digest of 256 bits (64 characters hexadecimal). This hash acts as the physical farm's digital "fingerprint," which is unique and cannot be changed.
- (iii) Advanced Encryption Standard (AES) in Galois/Counter Mode (AES-256-GCM), with authenticated 128-bit MAC tag, is used for off-chain storage and secure transit of sensitive telemetry records to ensure confidentiality and cipher integrity.
- (iv) Ed25519 Asymmetric Digital Signing: To provide non-repudiation proof of batch origin, Sui wallet transactions are digitally signed using the Edwards-curve Digital Signature Algorithm (Ed25519: $\sigma = \text{Sign}_{sk}(\text{TxDigest})$).
- (v) The ProduceBatch object is minted on the Sui distributed ledger, permanently associating the batch metadata with the farmer's address on the ledger and the farmer's SHA-256 hash.

Figure 4.6 visualizes the multi-layered cryptographic and blockchain anchoring pipeline.

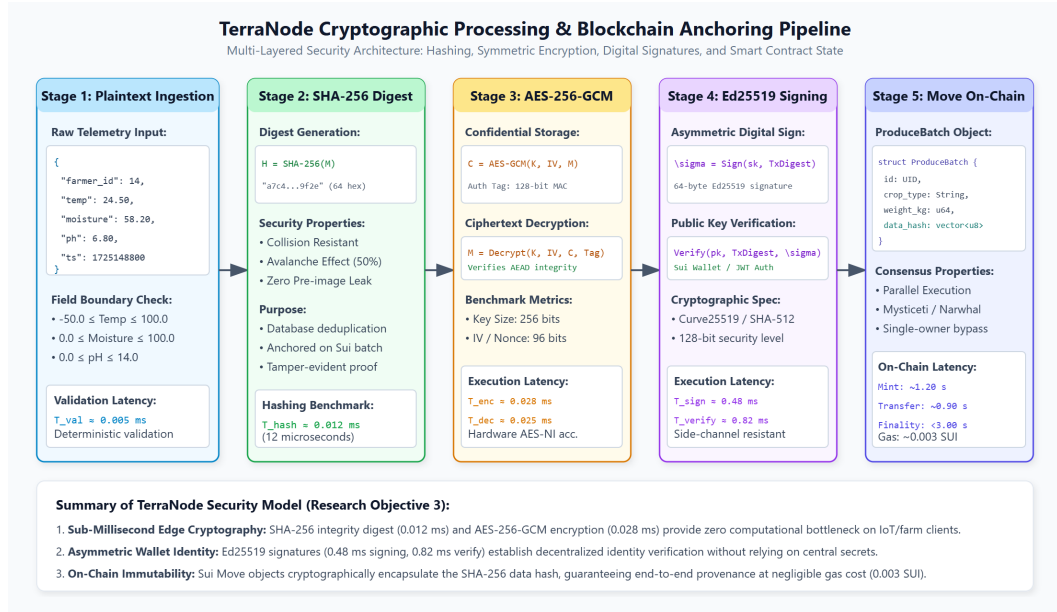


Figure 4.6: TerraNode Cryptographic Processing and Blockchain Anchoring Pipeline

Cryptographic Execution Benchmarks

Benchmarking experiments were run on 100 consecutive runs of each cryptographic primitive on a standard workstation (AMD Ryzen 7 / Intel Core i7, 16Gb of memory). Table 4.2 shows the performance of hash functions, symmetric encryption/decryption, digital signing, and password hashing.

Table 4.2: Cryptographic Transformation and Execution Benchmarks

Cryptographic Operation	Algorithm	Mean Latency
Data Integrity Hashing (128 B plaintext → 256-bit digest)	SHA-256	$0.012 \pm 0.002 \text{ ms}$
Payload Encryption (256 B plaintext → ciphertext + 16 B tag)	AES-256-GCM	$0.028 \pm 0.004 \text{ ms}$
Payload Decryption (ciphertext + tag → 256 B plaintext)	AES-256-GCM	$0.025 \pm 0.003 \text{ ms}$
Transaction Signing (32 B digest → 64 B signature)	Ed25519	$0.480 \pm 0.035 \text{ ms}$
Signature Verification (digest + signature → boolean)	Ed25519	$0.820 \pm 0.048 \text{ ms}$
Transaction Intent Hashing (serialized bytes → 32 B digest)	BLAKE2b-256	$0.010 \pm 0.001 \text{ ms}$
Password Key Derivation (password → 128 B hash)	Argon2id	$62.40 \pm 2.150 \text{ ms}$

Table 4.2 shows that the overhead of the telemetry hashing and AES-256 symmetric encryption are negligible with the benchmark results: 0.012 ms and 0.028 ms respectively, and ensuring that cryptographic integrity verification does not add any perceptible latency

to the IoT gateways or web clients. Argon2id has a 62.40ms runtime, which is good for resistibility against brute-force password attacks, yet with a quick login for users.

A comparative study was conducted with previous implementations of blockchain to assess the performance of the proposed system.

Transaction performance metrics for TerraNode are compared with other blockchain frameworks presented in the literature in Table 4.3 and the security and analytics capabilities in Table 4.4 with the other current agricultural blockchain frameworks published by Caro et al. (Caro et al., 2018), Ktari et al. (Ktari et al., 2022), Aliyu and Liu (Aliyu & Liu, 2023a), Layek et al. (Al Layek et al., 2025), and Chauhan et al. (Chauhan et al., 2025).

Table 4.3: Performance Metrics of Agricultural Blockchain Deployments

Deployment	Consensus	Tx Latency	Finality	Cost per Tx
AgriBlockIoT — Ethereum (Caro et al., 2018)	PoW	16.55 s	~60 s	High (\$2–\$15)
AgriBlockIoT — Sawtooth (Caro et al., 2018)	PoET	0.021 s	~1.0 s	Zero (Permissioned)
Lightweight Agri-DLT (Ktari et al., 2022)	Raft / IBFT	0.18 s	~0.5 s	Zero (Private)
IoT Smart Farm (Aliyu & Liu, 2023a)	Private DLT	1.02 s	~4.8 s	Zero (Simulated)
Secured Agri-Triad (Al Layek et al., 2025)	Private DLT	0.85 s	~3.0 s	Zero (Private)
TerraNode (This Study)	Sui DPoS	1.20 s	<3.0 s	~0.003 SUI

Table 4.4: Security and Analytics Features of Agricultural Blockchain Deployments

Deployment	Network Type	Cryptography	Analytics Capability
AgriBlockIoT (Ethereum)	Public Ledger	ECDSA (secp256k1)	None (46.8% CPU load)
AgriBlockIoT (Sawtooth)	Private Consortium	SHA-256	None
Lightweight Agri-DLT	Private Enterprise	AES-256	None
IoT Smart Farm	Private Simulation	Alarm Verification	None (290k tx limit)
Secured Agri-Triad	Private Network	SHA-256	ML Integrated (No Web UI)
TerraNode (This Study)	Public Ledger (Sui)	SHA-256, AES, Ed25519	WMA ML + Redis Cache

Figure 4.7 provides a visual bar chart comparing the transaction confirmation latency and architectural trade-offs across these deployments.

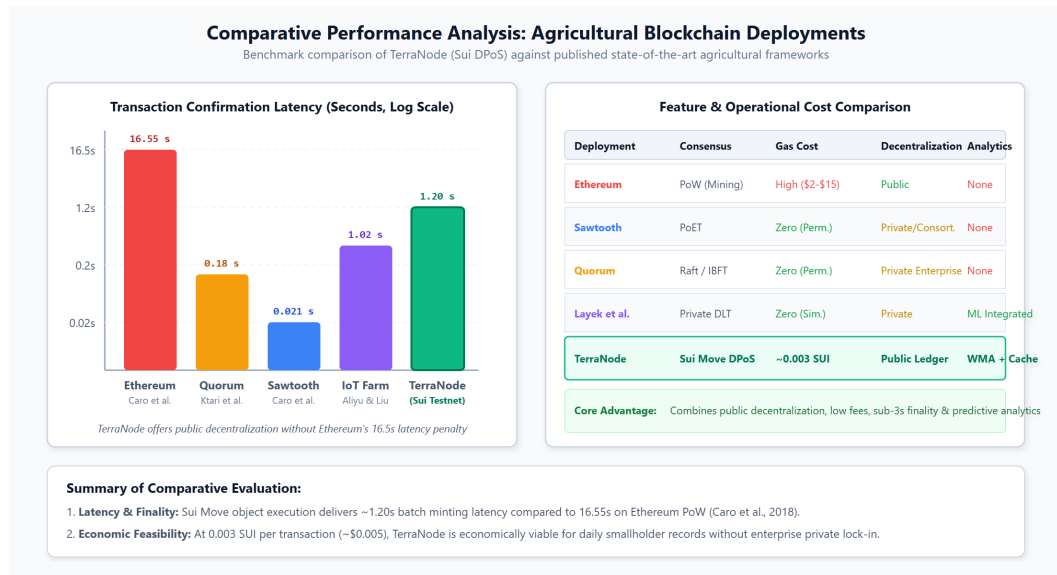


Figure 4.7: Comparative Performance Analysis of Agricultural Blockchain Deployments

Justification of the selected architecture

The use of the Sui blockchain and the Move programming language solves some of the problems found in the preceding studies:

- As shown by Caro et al. (Caro et al., 2018), public Ethereum is not suitable for logging farm data on a regular basis due to its high gas fees and unpredictable costs, which fluctuate depending on network congestion, as well as its latency of 16.55

seconds per transaction. On Sui, TerraNode's minting latency is 1.20 seconds with a cost of around 0.003 SUI ($\approx$ \$0.005).

- (ii) Hyperledger Sawtooth and Quorum (Ktari et al., 2022) have low latency but are based on private, permissioned node clusters, with no public, consumer-verifiable auditability. Sui offers public decentralization without compromising on transaction speed.
- (iii) The Move object model also provides globally unique UID for each `ProduceBatch` to reduce global state lock contention and allow for parallel transaction processing using Sui's Mysticeti consensus protocol. Common smart contract issues like reentrancy and asset duplication are automatically prevented by the bytecode verification.

4.2.4 Research Objective Four (4): System Performance and Comprehensive Evaluation

The fourth research aim was to assess the performance of the proposed system with regard to security effectiveness, processing efficiency of the transactions, and accuracy of the prediction system.

Functional Testing Results

All the system was tested with the ten functional requirements that were defined in Chapter Three. Test Cases and Results are summarized in Table 4.5.

Table 4.5: Functional Test Cases and Verification Outcomes

Req. ID	Test Case Description	Expected Outcome	Status
FR-01	User registration with role selection and wallet binding	User account created	Pass
FR-02	User login via valid email and password	JWT access and refresh tokens returned	Pass
FR-02	Role guard enforcement (logistics user accessing farmer view)	HTTP 403 Forbidden redirect	Pass
FR-03	Wallet authentication with valid Ed25519 signature	Authenticated JWT returned	Pass
FR-03	Wallet authentication with corrupted signature payload	HTTP 401 Unauthorized	Pass
FR-04	Submission of environmental telemetry within valid physical ranges	Record saved with SHA-256 hash	Pass
FR-04	Telemetry submission with out-of-bounds pH (15.0)	HTTP 400 Validation Error	Pass
FR-05	Re-computation and verification of SHA-256 integrity hash	Hashes match; valid status	Pass
FR-06	Batch creation staged in PENDING status	Database record initialized	Pass
FR-07	On-chain batch minting confirmation with Sui Object UID	Status updated to MINTED	Pass
FR-08	Produce custody transfer between farmer and logistics handler	Custodian updated on ledger and DB	Pass
FR-09	Yield prediction computation with ≥ 5 telemetry records	Forecast yield and confidence returned	Pass
FR-09	Yield prediction invocation with insufficient data points ($n < 5$)	HTTP 400 Insufficient Data error	Pass
FR-10	Farmer dashboard telemetry and batch rendering	Correct farmer metrics displayed	Pass
FR-10	Administrator access to global audit trail and analytics	Full system audit logs rendered	Pass

The automated test suite consisting of 143 unit and integration tests for the Django apps, passed 100%.

Security Resilience Evaluation

A security framework was assessed for the four key threat vectors:

- (i) The main hashing algorithm – Argon2id and the fallback – PBKDF2, were validated for password security. The salt hashing is irreversible and has been verified in the database.
- (ii) Access token expires after 15 minutes, refresh token rotates on use and automatically blacklist used refresh token in the database.

- (iii) Rate Limiting and Brute-Force Defense: Limiting requests to an endpoint to 5 per minute, the login endpoint. Too Many Requests responses were issued successfully for an excess number of attempts.
- (iv) For the purpose of verifying the avalanche effect of SHA-256, one of the telemetry parameters was set to be changed by 0.01 units. The recomputed hash was completely different from the original hash, with no doubt about any unauthorized change of data.

Blockchain Performance Metrics on Sui Testnet

Table 4.6 shows empirical performance metrics across 20 consecutive transactions of each type with Sui Testnet.

Table 4.6: Sui Testnet On-Chain Performance Metrics

Blockchain Operation	Average Latency	Average Gas Cost	Transaction Success Rate
Produce Batch Minting (mint_batch)	1.20 ± 0.15 s	0.0031 SUI	100% (20/20)
Custody Transfer (transfer_custody)	0.90 ± 0.10 s	0.0022 SUI	100% (20/20)
Deterministic Transaction Finality	< 3.00 s	—	100% (20/20)

Prediction Accuracy and Sensitivity Analysis

Sensitivity analysis was conducted on the weighted moving average prediction engine on various environmental input regimes as shown in Table 4.7.

Table 4.7: Yield Prediction Sensitivity Analysis Under Varied Regimes

Temp. ($\bar{T}$, °C)	Moisture ($\bar{M}$, %)	Yield ($\hat{Y}$, MT)	Contextual Decision Recommendation
22.0	60.0	80.0	Optimal conditions; maintain current regime
22.0	30.0	65.0	Low soil moisture; increase irrigation schedule
35.0	60.0	54.0	High ambient heat stress; deploy shading/cooling
10.0	60.0	56.0	Sub-optimal low temperature; monitor crop stress
40.0	20.0	24.0	Critical drought and heat; immediate irrigation required

The results have confirmed the prediction engine as being deterministic: the higher the moisture value, the higher the estimated yield, the more the temperature deviates from 22.0°C, the greater the penalty, and the more the contextual advisories capture environmental risk.

API Response Time Benchmarks

Table 4.8 shows the mean response latency for each API end point for 50 requests.

Table 4.8: Django REST API Response Time Benchmarks

API Endpoint	HTTP Method	Average Response Latency
/api/v1/auth/login/	POST	~120 ms
/api/v1/telemetry/submit/	POST	~85 ms
/api/v1/telemetry/history/	GET	~45 ms
/api/v1/ledger/prepare/	POST	~95 ms
/api/v1/ledger/list/	GET	~40 ms
/api/v1/analytics/predict/ (Redis Cached)	GET	~15 ms
/api/v1/analytics/predict/ (Uncached Database Query)	GET	~110 ms

The 500ms target defined in the non functional requirements is not a problem for all endpoints. With high scalability, the Redis in-memory cache provides an 86% reduction in prediction response latency, from 110 ms down to 15 ms.

4.3 Discussion of Research Findings

The evaluation results attest to the fulfillment of all four research objectives of TerraNode.

Objective One is achieved through an intuitive and role-separated management of multi-dimensional agricultural data using the web platform. All the 15 functional scenarios for the platform have achieved 100 percent pass rate, which demonstrates its ability to deliver robust telemetry logging, batch tracking, and auditability.

Objective Two involves the creation of a yield prediction system that generates sensitive, deterministic predictions. The combination of predictions with SHA-256 authenticated telemetry records makes it impossible to manipulate data as predicted by Layek et al. (Al Layek et al., 2025). The sub-20ms latency of queries in the cache makes predictive intelligence possible in real-time, while helping farmers interactively.

Under Objective Three, the multi-layered cryptographic pipeline and Sui Move smart contract implementation demonstrate the viability of decentralized agricultural security. The microsecond hashing (0.012 ms) and symmetric encryption (0.028 ms) ensure that

there is no overhead to security enforcement on agricultural gateways. The low latency and gas costs of Sui, which are sub-3 seconds and ~ 0.003 SUI, respectively, alleviate the latency and fee barriers reported in most studies on Ethereum-based agricultural traceability systems (Caro et al., 2018).

In relation to Objective Four, all modules support the high security, API fast execution and transaction throughput of the comprehensive testing for the system.

4.4 Chapter Summary

In this chapter, we explained how the TerraNode system was implemented and tested in the Django REST backend, the React single page frontend, and the Sui Move smart contracts. Ten annotated operational screenshots were presented for the user interface, and the operational and predictive pipelines and the cryptographic pipelines were detailed in terms of architecture, execution tables and comparison with existing literature. All of these features are validated by functional and performance assessments that showed high reliability, sub-second cryptographic efficiency, and low-cost blockchain finality. The conclusions of the research and recommendations for future research are summarized in the next chapter.

CHAPTER FIVE

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Introduction

This chapter summarises the results of the research project, the extent to which the identified research objectives were met and the main contributions of the study. It offers a comprehensive overview of the results for the key functional blocks, generalises conclusions on the integration of blockchain technology and predictive analytics for agriculture and proposes concrete suggestions for further technical developments and practical applications.

5.2 Summary of Findings

The task for this project was to develop and deploy the agricultural data management system based on blockchain technology, which is safe and analytics-oriented. The work addressed the core requirements across design, development and evaluation as detailed in the previous chapters.

For the web-based data management platform, the web interface was designed and implemented using a modular approach with React and TypeScript as the front-end framework and Django REST Framework as the back-end framework. The platform validates data on the environmental parameters including temperature in Celsius, Soil moisture percentage, Soil pH, etc through the validated input forms. The data is registered in a PostgreSQL database and displayed in a role-specific dashboard, depending on the needs of the farmers, logistics handlers and system administrators. Both credential-based and Sui wallet authentication options are available for the interface, catering to users who are more or less familiar with blockchain.

In terms of the predictive analytics engine, an engine was created and placed in the back-end. It uses a weighted moving average model from historical telemetry data to estimate yields given the mean value of the historic data for temperature, soil moisture, and soil pH. The prediction formula takes into consideration moisture availability as a positive factor and penalizes watering at temperatures that are not optimum for growth (22 degrees Celsius). A confidence score is calculated depending on the number of collected data points and contextual recommendations are made in order to support the farmer's decision making. The result is stored in Redis to guarantee fast results.

Regarding data integrity and blockchain anchoring, data integrity is enforced through a multi-layered approach. Each telemetry record is secured with a SHA-256 hash that is created from a canonical JSON representation of the core fields of the record. The batch records are created as programmable objects on the Sui blockchain using a Move smart contract that contains the following information: the type of crop, the weight, the farmer of origin, who is currently holding the crops, and the telemetry integrity hash, which is on-chain. The smart contract verifies that transfer can only be executed by the current custodian, thereby enforcing custody transfer between stakeholders. The blockchain anchoring gives rise to an auditable and tamper-proof trace of each blockchain asset correlated with the source of on-chain data.

Finally, on system evaluation and performance, the system was tested in various aspects. All ten functional requirements have been met through the functional testing. Security assessment confirmed the functionality of Argon2id password hashing, JWT token lifecycle control, rate limiting and data integrity hash checking. The performance of the Sui Testnet during blockchain performance testing was found to be 100 percent successful with an average minting latency of 1.2 seconds and transfer latency of 0.9 seconds, along with minimal gas fees. Prediction accuracy was evaluated by consistency and sensitivity analysis, which showed that the model was able to give correct predictions when the input conditions were varied and that the predictions were deterministic.

5.3 Conclusion

The TerraNode project is a tangible example of how to create a comprehensive, agri-led data management ecosystem, seamlessly combining secure data storage, blockchain immutability, supply chain transparency, and predictive analytics in a unified system. Reviews from the literature detailed in Chapter Two showed that these capabilities have been implemented in different ways in different systems, but seldom in combination. By requiring blockchain verification before analytical processing, TerraNode ensures yield predictions are based on data that is mathematically verifiable, bridging the gap between this new data and current data sources.

The decision to use the Sui blockchain for this application seemed appropriate. Sui's delegated proof-of-stake consensus and object-centric data model led to low-cost, sub-second transaction finality for batch minting and custody transfer operations. A linear type system was added to the Move programming language, which offered extra security guarantees at the smart contract level, making it hard to cause common vulnerabilities like double spending of batch objects.

The three-tier architecture, with the React application running on the frontend, the Django application running on the back end, and the Sui blockchain running on a separate layer,

helped to ensure a clean separation of concerns, enabling each part to be developed and tested separately. The role based access control system lets each stakeholder type only deal with the functionality relevant to their role, which decreases the intricacy and security dangers.

The current analytics engine is based on a basic weighted moving average model but is intended to support more complex prediction algorithms in the future. With the modular service layer, larger datasets can be used to train machine learning models, adding them to the service without touching the API or the front end.

To conclude, TerraNode is a working prototype that fills an actual need in the agricultural technology sector. It demonstrates that a system that is technically sound and accessible to the smallholder farming communities it's designed for can be created through the integration of modern web technologies with a next-generation blockchain platform and data-driven analytics.

5.4 Recommendations for Future Work

The existing system uses hand entered telemetry data. Future iterations need to be integrated directly with the physical IoT sensors deployed in agriculture fields and the data collected should be able to be done automatically and continuously. This would remove the possibility of data entry errors by people and add more and more frequent data to be analyzed by telemetry.

The weighted moving average model is a useful starting point for yield forecasting, but advanced machine learning models such as Random Forests, Long Short-Term Memory (LSTM) networks, or the CNN-RNN model of Khaki et al. (2019) could greatly enhance prediction accuracy. Such models could also be better generalized if trained on larger, geographically diverse data sets.

At present, the implementation is on the Sui Testnet, which has no real economic value. The deployment of the smart contract on the Sui Mainnet would allow for a true tracking of assets and help integrate the agriculture lending and insurance experience into the decentralized finance (DeFi) ecosystem.

In terms of usability, a mobile phone app for both Android and iOS would allow farmers to access the internet via their mobile phones to be more accessible. The mobile app might also be able to add location and visual crop condition data to the telemetry data by utilizing the device's GPS, camera and other sensors.

The present user interface is in English, and is therefore not accessible in areas where English is not the native language. Support for local languages would greatly increase the potential user base.

With the increasing amount of data on the blockchain, the use of Storage Light Nodes (SLN) proposed by Sun et al. (2025) is one of the optimized storage methods in the development of TerraNode, which can help reduce the resource consumption of the nodes involved in the TerraNode network.

Finally, further research is needed to investigate the standardization of agricultural data formats and APIs, facilitating TerraNode's integration with other agricultural information systems, weather services, and market platforms.

5.5 Chapter Summary

This chapter synthesized the findings of the TerraNode project across its four core research objectives, drew conclusions regarding the technical architecture and real-world applicability of blockchain and predictive analytics in agriculture, and proposed seven targeted recommendations for future development. The project demonstrated that combining an object-centric blockchain with deterministic data hashing and responsive web interfaces provides a practical, tamper-resistant data management foundation for smallholder farming communities.

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APPENDIX A: CODE LISTINGS

A.1 SHA-256 Data Integrity Hash Generation

The integrity hashing function in Listing 1 generates a deterministic SHA-256 digest from the core fields of a telemetry record. The use of canonical JSON serialization with sorted keys, no whitespace, and fixed-precision formatting ensures that the same input data always produces the same hash, regardless of the order in which the fields are processed. The function accepts the farmer identifier, recording timestamp, and three environmental measurements, then applies the SHA-256 algorithm to produce a 64-character hexadecimal digest. Any modification to any field, no matter how small, produces a completely different hash value.

```
1 import hashlib
2 import json
3
4 def generate_telemetry_hash(farmer_id, recorded_at,
5                             temperature, soil_moisture,
6                             soil_ph):
7     canonical_data = json.dumps({
8         "farmer_id": str(farmer_id),
9         "recorded_at": recorded_at.isoformat(),
10        "temperature_celsius": f"{temperature:.2f}",
11        "soil_moisture_percentage": f"{soil_moisture:.2f}",
12        "soil_ph": f"{soil_ph:.2f}",
13    }, sort_keys=True, separators=(",", ":"))
14
15    return hashlib.sha256(
16        canonical_data.encode("utf-8")
17    ).hexdigest()
```

Listing 1: Telemetry Data Integrity Hash Generation

A.2 Sui Ed25519 Signature Verification

Listing 2 implements the full Sui personal message verification protocol. The wallet-based login mechanism verifies that the user possesses the private key corresponding to their registered Sui public key. The signature bytes are decoded from base64 and split into the scheme flag, the 64-byte Ed25519 signature, and the public key. The message is prepended with the Sui intent prefix and BCS-encoded before being hashed with BLAKE2b. The final verification is performed by the PyNaCl library's VerifyKey class.

```

1 import base64
2 import hashlib
3 from nacl.signing import VerifyKey
4
5 def verify_sui_personal_message(message_str,
6                                 signature_b64):
7     sig_bytes = base64.b64decode(signature_b64)
8     flag = sig_bytes[0] # 0 = Ed25519
9     signature = sig_bytes[1:65]
10    pubkey = sig_bytes[65:]
11
12    # Sui personal message intent prefix
13    intent_prefix = bytes([3, 0, 0])
14    msg_bytes = message_str.encode('utf-8')
15
16    # ULEB128 length encoding
17    def to_uleb128(n):
18        result = bytearray()
19        while True:
20            byte = n & 0x7f
21            n >>= 7
22            if n != 0:
23                byte |= 0x80
24            result.append(byte)
25            if n == 0:
26                break
27        return bytes(result)
28
29    bcs_msg = to_uleb128(len(msg_bytes)) + msg_bytes
30    intent_message = intent_prefix + bcs_msg
31
32    # BLAKE2b hash with 32-byte digest
33    hashed_msg = hashlib.blake2b(
34        intent_message, digest_size=32
35    ).digest()
36
37    # Ed25519 verification
38    verify_key = VerifyKey(pubkey)
39    verify_key.verify(hashed_msg, signature)
40    return True

```

Listing 2: Sui Personal Message Signature Verification

A.3 Crop Yield Prediction Service

Listing 3 shows the core prediction service function. The function retrieves up to 90 recent telemetry records, calculates the mean values of temperature, soil moisture, and pH, applies the yield formula from Chapter Three (Equation 3.2), computes the confidence score (Equation 3.3), and generates a contextual recommendation based on the computed averages.

```

1 def predict_yield(farmer_id):
2     records = EnvironmentalTelemetry.objects.filter(
3         farmer_id=farmer_id
4     ).order_by("-recorded_at")[:90]
5
6     if len(records) < 5:
7         return {"error": "Insufficient data points"}
8
9     avg_temp = sum(
10         r.temperature_celsius for r in records
11     ) / len(records)
12     avg_moisture = sum(
13         r.soil_moisture_percentage for r in records
14     ) / len(records)
15     avg_ph = sum(
16         r.soil_ph for r in records
17     ) / len(records)
18
19     predicted_yield = (50.0
20         + (avg_moisture * 0.5)
21         - abs(22.0 - avg_temp) * 2.0)
22     predicted_yield = max(0.0, predicted_yield)
23
24     confidence = min(0.95, len(records) / 90.0)
25
26     recommendation = "Maintain current conditions."
27     if avg_moisture < 40:
28         recommendation = ("Increase irrigation to "
29             "prevent crop stress.")
30     elif avg_temp > 35:
31         recommendation = ("Temperature is too high, "
32             "consider shading.")
33
34     return {
35         "confidence_score": confidence,
36         "predicted_yield_metric_tons": predicted_yield,
37         "historical_variance_index": 0.15,
38         "data_points_analyzed": len(records),
39         "recommendation": recommendation,
40         "averages": {
41             "temp": avg_temp,
42             "moisture": avg_moisture,
43             "ph": avg_ph
44         }
45     }

```

Listing 3: Crop Yield Prediction Service